



Global bifurcation for a predator-prey system with nonlinear boundary-mediated dispersal [★]

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ABSTRACT

This paper presents a complete bifurcation analysis of steady-state solutions for a predator-prey reaction-diffusion-advection system featuring density-dependent boundary conditions. By developing a novel synthesis of local and global bifurcation theories, we characterize the full solution structure emerging from semi-trivial states, revealing how nonlinear boundary terms qualitatively alter system dynamics. Our main results establish: (i) precise conditions for coexistence state bifurcations that coincide with stability transitions of semi-trivial solutions; (ii) complete global continuation of solution branches showing either unbounded persistence or novel connection patterns; and (iii) quantitative criteria for stability thresholds governed by boundary-mediated feedback. The analysis overcomes significant technical challenges through sharp a priori estimates and careful treatment of non-self-adjoint operators arising from the nonlinear boundary conditions. These theoretical advances provide mechanistic explanations for edge effects in ecological systems while developing mathematical tools applicable to broader pattern formation problems. The work opens new directions for studying nonlinear boundary phenomena in biological systems, with natural extensions to multi-species communities and chemotaxis models.

1. Introduction

Predator-prey dynamics have served as a fundamental paradigm in mathematical ecology since the pioneering work of Lotka and Volterra [1–3], with reaction-diffusion systems emerging as essential tools for understanding spatiotemporal population patterns. While these models have provided profound insights into species coexistence, traditional formulations often neglect the crucial role of boundary effects in shaping dispersal behaviors. In natural ecosystems, habitat boundaries function as ecological interfaces where density-dependent emigration and immigration processes create complex feedback mechanisms. These nonlinear boundary conditions introduce significant mathematical challenges while fundamentally altering system dynamics, necessitating novel analytical approaches to unravel their intricate nature.

Recent advances in nonlinear boundary value problems [4–10] have demonstrated their profound impact on pattern formation and stability properties. Foundational contributions to this field can be found in the classical works [11,12]. Density-dependent dispersal at boundaries induces rich bifurcation phenomena that differ qualitatively from conventional models. Although local bifurcation analysis has been developed for various ecological systems [13–16], the global bifurcation structure for coupled systems with nonlinear

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boundary conditions remains largely unexplored. This knowledge gap limits our ability to understand how spatial heterogeneity and boundary-mediated processes jointly influence coexistence patterns in ecological communities.

The study of scalar equations with nonlinear boundary conditions has yielded significant theoretical insights. Previous work has established fundamental results on well-posedness [17–20], blow-up behavior [21–25], and boundary layer solutions [26–29]. Bifurcation methods have proven particularly effective for analyzing steady-state solutions in various contexts [30–44]. A notable contribution by Umezū [39] examined the logistic equation with nonlinear flux boundary conditions:

$$\begin{cases} \frac{\partial u}{\partial t} = d\Delta u + u(m(x) - u), & \text{in } \Omega \times \mathbb{R}_+, \\ d \frac{\partial u}{\partial \mathbf{n}} = q(x)u^2, & \text{on } \partial\Omega \times \mathbb{R}_+. \end{cases} \quad (1.1)$$

Here, $\mathbb{R}_+ = (0, \infty)$ and the boundary flux depends quadratically on population density. Using the method developed by Lyapunov and Schmidt, along with a scaling argument, the existence of a bifurcation branch of positive steady-state solutions emanating from trivial steady-state solutions is established, along with their asymptotic behavior and stability. However, the global bifurcation approach employed in [39] is not universally applicable and cannot be extended to predator-prey systems, highlighting the need for more versatile analytical techniques to address such complex interactions.

Subsequent work by Liu and Shi [4] developed a more general framework for reaction-diffusion systems with nonlinear boundary conditions:

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u + \lambda s(x)f(u), & \text{in } \Omega \times \mathbb{R}_+, \\ \frac{\partial u}{\partial \mathbf{n}} = \lambda r(x)g(u), & \text{on } \partial\Omega \times \mathbb{R}_+, \end{cases} \quad (1.2)$$

where the reaction term $f(u)$ stands for a general growth rate of species u , and $g(u)$ denotes the nonlinear boundary condition. Liu and Shi [4] explored the bifurcation problem for nontrivial steady-state solutions of model (1.2) by applying abstract local bifurcation theorems from [45–48] and global bifurcation theory from [49,50]. Recently Li, Dai and Zou [14] employed a similar approach to establish a general bifurcation result for a reaction-diffusion-convection model with nonlinear boundary conditions, extending the methodology used for the non-advection model (1.2).

To the best of our knowledge, there are currently only a few competing models for reaction-(advection)-diffusion systems with nonlinear boundary conditions, as seen in [6,51,52]. In contrast, predator-prey models with linear boundary conditions have been extensively studied [53–58]. In particular, Li and Wang [59] applied unilateral global bifurcation theory, as developed by Cintra et al. [60], to provide a comprehensive analysis of the global bifurcation structure of coexistence steady states for the following predator-prey system with prey taxis under homogeneous Dirichlet boundary conditions:

$$\begin{cases} \frac{\partial u}{\partial t} = \nabla \cdot (d(v)\nabla u - u\chi(v)\nabla v) + \lambda u - u^2 + \gamma uF(v), & \text{in } \Omega \times \mathbb{R}_+, \\ \frac{\partial v}{\partial t} = D\Delta v + \mu v - v^2 - uF(v), & \text{in } \Omega \times \mathbb{R}_+, \\ u = v = 0, & \text{on } \partial\Omega \times \mathbb{R}_+, \end{cases} \quad (1.3)$$

where constants $\lambda \in \mathbb{R}$ and $\mu \in \mathbb{R}_+$ denote the growth rates of each species; constant $\gamma > 0$ accounts for the intrinsic predation rate; nonnegative function $F(v)$ represents the functional response of the predator; positive constant D is the diffusion rate of the prey; the term $-\nabla u(\chi(v)\nabla v)$ describes the prey-taxis with coefficient $\chi(v)$.

The unilateral global bifurcation theory developed by Cintra et al. [60], which builds upon Theorem 6.4.3 of [61], is specifically tailored for systems with linear parameter dependence and Dirichlet boundary conditions. As such, it cannot be directly applied to nonlinear scenarios, particularly those involving nonlinear boundary conditions. Furthermore, the unilateral theorems originally proposed by Rabinowitz [50] have been shown to contain significant flaws, as critically discussed by López-Gómez [61] and Shi and Wang [49], with Dancer's counterexample [62] highlighting essential gaps in their proofs. Moreover, although the local bifurcation theorem of Crandall and Rabinowitz can be successfully applied, it follows from Corollary 5.5 of López-Gómez and Mora-Corral [63] that the component bifurcating from a simple eigenvalue must satisfy Rabinowitz's global alternative [50]. However, this global result does not guarantee that the subcomponent consisting of coexistence states will itself satisfy Rabinowitz's global alternative.

Fortunately, Shi and Wang [49] generalized Theorem 6.4.3 in [61] to address nonlinear boundary problems for quasilinear systems and cases where parameters exhibit nonlinear dependence. However, verifying the differentiability of the norm in this framework remains challenging. This condition was later refined by López-Gómez [64] into a more tractable compact inclusion form, significantly advancing the applicability of the theory to complex nonlinear systems.

In this paper, we focus on the following Lotka-Volterra predator-prey system with nonlinear boundary conditions

$$\begin{cases} \frac{\partial u_1}{\partial t} = d_1 \nabla \cdot [\nabla u_1 - \alpha_1 u_1 \nabla A_1(x)] + u_1[m_1(x) - u_1 - au_2] & \text{in } \Omega \times \mathbb{R}_+, \\ \frac{\partial u_2}{\partial t} = d_2 \nabla \cdot [\nabla u_2 - \alpha_2 u_2 \nabla A_2(x)] + u_2[m_2(x) - u_2 + bu_1] & \text{in } \Omega \times \mathbb{R}_+, \\ d_1 \left[\frac{\partial u_1}{\partial \mathbf{n}} - \alpha_1 u_1 \frac{\partial A_1(x)}{\partial \mathbf{n}} \right] = q_1(x)u_1^2 & \text{on } \partial\Omega \times \mathbb{R}_+, \\ d_2 \left[\frac{\partial u_2}{\partial \mathbf{n}} - \alpha_2 u_2 \frac{\partial A_2(x)}{\partial \mathbf{n}} \right] = q_2(x)u_2^2 & \text{on } \partial\Omega \times \mathbb{R}_+, \end{cases} \quad (1.4)$$

where $A_i \in C^{2+\sigma}(\overline{\Omega})$, $i = 1, 2$ for some $\sigma \in (0, 1)$, u_1 and u_2 represent the population densities of the prey and predator at location x and time t , respectively; $d_i > 0$ is the diffusion rate, α_i be the advection rate, $a, b > 0$ are the conversion rate, $m_1(x)$ and $m_2(x)$ represent their intrinsic growth rates, ∇ is the gradient operator on $\mathbb{R}^N$, the habitat Ω is a bounded region in $\mathbb{R}^N$ with $C^{2+\sigma}$ -smooth boundary $\partial\Omega$ for some $\sigma \in (0, 1)$, $\mathbf{n}$ is the outward unit normal vector on the boundary $\partial\Omega$.

Here, $m_1(x)$ and $m_2(x)$ can take positive values in favorable habitats and negative values in unfavorable ones. In particular, if $m_2(x) < 0$ on Ω , it implies that the predator relies solely on the prey u for sustenance, and $m_2(x)$ represents the predator's death rate in the absence of prey. Unlike classical models with passive dispersal, our nonlinear boundary conditions model active density-dependent edge behaviors with profound ecological consequences. Mathematically, these terms introduce a new stabilizing mechanism, may create new bifurcation scenarios not observed in linear boundary systems, and also fundamentally alter the system's analytical properties by breaking standard comparison principles due to the quadratic nonlinearity. Biologically, the boundary terms can drive emergent ecological phenomena and favor specific spatial distributions. Throughout the paper, we always assume that

(A1) $m_i \in C^\sigma(\overline{\Omega})$ for some $\sigma \in (0, 1)$ is nonconstant and satisfies $\int_{\Omega} m_i(x) dx > 0$ and $\min_{\Omega} m_i < 0$, $i = 1, 2$.

(A2) $q_i \in C^\sigma(\partial\Omega)$ for some $\sigma \in (0, 1)$ and satisfies $q_i(x) \leq 0$ for all $x \in \partial\Omega$, $i = 1, 2$.

This paper is structured as follows. In Section 2, we state our main results. Section 3 lays the mathematical groundwork by establishing fundamental results for eigenvalue problems with indefinite weights, which provide the necessary theoretical foundation for our subsequent analysis of single-species population models. These preliminary results enable us to systematically investigate the existence and stability properties of semi-trivial steady-state solutions in Section 4, where we develop crucial insights into the baseline dynamics of system (1.4). The analytical framework is further strengthened in Section 5 through the derivation of a priori $C(\Omega)$ -estimates for positive steady-state solutions, an essential technical tool that underpins our global bifurcation analysis. Building on these estimates, Section 6 applies the Crandall-Rabinowitz local bifurcation theory to rigorously prove the existence of solution branches originating from semi-trivial states. This local analysis is then extended to a global context in Section 7 through the application of unilateral global bifurcation theory [64], which reveals the complete solution structure and interconnection patterns between different equilibrium states. We conclude in Section 8 with a comprehensive discussion of our main theoretical findings, their biological implications, and several promising directions for future research.

To facilitate reading, we employ standard functional analysis notation throughout: $L^p(\Omega)$ ($p \geq 1$) denotes the Lebesgue space of p -integrable functions on Ω , while $W^{k,p}(\Omega)$ ($k \geq 0$) represents the Sobolev space containing L^p -functions whose weak derivatives up to order k also belong to $L^p(\Omega)$.

2. Main results

The non-self-adjoint nature of the advection operator $\nabla \cdot [\nabla w - \alpha w \nabla A(x)]$ necessitates a transformation to facilitate analysis. By introducing the variables $u = e^{-\alpha_1 A_1} u_1$ and $v = e^{-\alpha_2 A_2} u_2$, we reformulate the original system (1.4) into a more tractable form

$$\begin{cases} \frac{\partial u}{\partial t} = d_1 e^{-\alpha_1 A_1} \nabla \cdot [e^{\alpha_1 A_1} \nabla u] + u[m_1 - e^{\alpha_1 A_1} u - a e^{\alpha_2 A_2} v] & \text{in } \Omega \times \mathbb{R}_+, \\ \frac{\partial v}{\partial t} = d_2 e^{-\alpha_2 A_2} \nabla \cdot [e^{\alpha_2 A_2} \nabla v] + v[m_2 - e^{\alpha_2 A_2} v + b e^{\alpha_1 A_1} u] & \text{in } \Omega \times \mathbb{R}_+, \\ d_1 \frac{\partial u}{\partial n} = q_1 e^{\alpha_1 A_1} u^2 & \text{on } \partial\Omega \times \mathbb{R}_+, \\ d_2 \frac{\partial v}{\partial n} = q_2 e^{\alpha_2 A_2} v^2 & \text{on } \partial\Omega \times \mathbb{R}_+, \end{cases} \quad (2.1)$$

where the nonlinear boundary terms $q_1 e^{\alpha_1 A_1} u^2$ and $q_2 e^{\alpha_2 A_2} v^2$ explicitly couple species densities with environmental gradients. This transformation reveals how advection along habitat features A_1 and A_2 interacts with density-dependent dispersal at boundaries.

Because semi-trivial solutions provide valuable insights into the structure of positive steady states, we begin by establishing existence results for these solutions of (2.1) and obtain the following result, whose complete proof appears in Section 4 (see Lemma 4.1 for existence results and Lemma 4.4 for stability analysis).

Theorem 2.1 (Existence of Semi-Trivial Solutions). *Under conditions (A1) and (A2), system (2.1) admits a semi-trivial steady-state solution $(u, v) = (\theta_{d_1, m_1, q_1}, 0)$ provided that*

(H1) $\int_{\Omega} e^{\alpha_1 A_1} m_1 dx \geq 0$, or $\int_{\Omega} e^{\alpha_1 A_1} m_1 dx < 0$ and $0 < d_1 < v_1(e^{\alpha_1 A_1} m_1)$,
and a semi-trivial steady-state solution $(u, v) = (0, \theta_{d_2, m_2, q_2})$ provided that

(H2) $\int_{\Omega} e^{\alpha_2 A_2} m_2 dx \geq 0$, or $\int_{\Omega} e^{\alpha_2 A_2} m_2 dx < 0$ and $0 < d_2 < v_1(e^{\alpha_2 A_2} m_2)$.

Here $v = v_1(e^{\alpha A} m)$ denotes the principal eigenvalue of the weighted eigenvalue problem:

$$-\nabla \cdot [e^{\alpha A} \nabla w] = e^{\alpha A} m(x) w \quad \text{in } \Omega, \quad \frac{\partial w}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

The above theorem establishes the existence conditions for semi-trivial steady states in our predator-prey system (2.1), where one species persists while the other is absent. Mathematically, the solutions $(\theta_{d_1, m_1, q_1}, 0)$ and $(0, \theta_{d_2, m_2, q_2})$ emerge through a delicate interplay between diffusion rates d_i , habitat quality m_i , and the advection potentials A_i , mediated by the principal eigenvalue v_i of a weighted elliptic operator. The critical threshold conditions in (H1)-(H2) reveal how the sign of weighted spatial averages of resource functions m_i ($i = 1, 2$) determines whether solutions exist for all diffusion rates or only below an eigenvalue-determined critical value. Biologically, these semi-trivial states represent crucial baseline scenarios where either predators or prey cannot establish themselves. The weighted integrals $\int_{\Omega} e^{\alpha_i A_i} m_i dx$ quantify net habitat quality after accounting for species' environmental preferences through A_i ,

where positive values indicate favorable conditions supporting population persistence regardless of mobility, while negative values restrict existence to low diffusion rates where organisms cannot escape unfavorable regions. The eigenvalue condition $v_1(e^{\alpha_i A_i} m_i)$ captures how environmental heterogeneity and directed movement (via $\alpha_i \nabla A_i$) interact to determine critical thresholds for population survival. These mathematical results thus provide precise conditions under which edge effects, modeled through nonlinear boundary terms q_i , permit single-species persistence in fragmented habitats.

In order to investigate the existence of positive steady-state solutions emerging from the semi-trivial steady-state solutions, we have to consider the linear stability of semi-trivial steady-state solutions $(\theta_{d_1, m_1, q_1}, 0)$ and $(0, \theta_{d_2, m_2, q_2})$ for $d_1 \in \mathbb{D}_1$ and $d_2 \in \mathbb{D}_2$, where

$$\mathbb{D}_j = \begin{cases} \mathbb{R}_+ & \text{if } \int_{\Omega} e^{\alpha_j A_j} m_j dx \geq 0, \\ (0, v_1(e^{\alpha_j A_j} m_j)) & \text{if } \int_{\Omega} e^{\alpha_j A_j} m_j dx < 0 \end{cases}$$

for $j = 1, 2$. In fact, we have the following results.

Theorem 2.2 (Stability of Semi-Trivial Solution $(\theta_{d_1, m_1, q_1}, 0)$). *Assume that (A1), (A2), and (H1) holds, then the stability of the semi-trivial steady-state solution $(\theta_{d_1, m_1, q_1}, 0)$ for system (2.1) is characterized as follows:*

- (i) $(\theta_{d_1, m_1, q_1}, 0)$ is unstable for all $d_2 > 0$ when $\int_{\Omega} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx \geq 0$ and $m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} < 0$ somewhere in Ω ;
- (ii) $(\theta_{d_1, m_1, q_1}, 0)$ loses its stability as d_2 decreases through the critical value $d_2 = d_2^0 \triangleq v_1(e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}))$ when

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx < 0. \quad (2.2)$$

The stability properties of the prey-only semi-trivial solution reveal fundamental mechanisms governing predator invasion dynamics in heterogeneous environments. Mathematically, the stability threshold is determined by a weighted integral condition involving both the predator's intrinsic growth rate m_2 and its functional response to prey density (through the term $b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}$), where the spatial averaging accounts for habitat heterogeneity through the environmental potential A_i . The critical diffusion rate d_2^0 marks a bifurcation point where the predator's ability to invade changes abruptly, with the principal eigenvalue v_1 encoding the combined effects of diffusion, advection, and spatial heterogeneity.

Biologically, these stability conditions demonstrate how nonlinear boundary-mediated dispersal creates two distinct ecological regimes. When the weighted average of predator resources is non-negative (Theorem 2.2(i)), the prey population can never be stabilized against predator invasion, regardless of predator mobility. Conversely, when resource availability falls below this threshold (Theorem 2.2(ii)), system (2.1) exhibits a critical predator dispersal rate d_2^0 below which invasion becomes possible. This reflects an ecological trade-off where low predator mobility prevents efficient prey tracking, while high mobility may lead to over-dispersion and population decline. The stability transition thus emerges from the interplay between predator foraging efficiency b , prey spatial distribution θ_{d_1, m_1, q_1} , and the predator's environmental response $\alpha_2 A_2$, with the nonlinear boundary conditions modifying these relationships through density-dependent edge effects.

Theorem 2.3 (Stability of Semi-Trivial Solution $(0, \theta_{d_2, m_2, q_2})$). *Assume that (A1), (A2), and (H2) holds, then the stability of the semi-trivial steady-state solution $(0, \theta_{d_2, m_2, q_2})$ of system (2.1) is characterized as follows:*

- (i) $(0, \theta_{d_2, m_2, q_2})$ is unstable for all $d_1 > 0$ when $\int_{\Omega} e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) dx \geq 0$,
- (ii) $(0, \theta_{d_2, m_2, q_2})$ loses its stability as d_1 decreases and passes through $d_1 = d_1^0 \triangleq v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}))$ when

$$\int_{\Omega} e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) dx < 0 \text{ and } m_1 - a e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2} > 0 \text{ somewhere in } \Omega, \quad (2.3)$$

- (iii) Assume further that

$$\int_{\Omega} e^{\alpha_2 A_2} m_2 dx < 0, \quad (2.4)$$

then there exists a threshold $d_2^* \in (0, v_1(e^{\alpha_2 A_2} m_2))$ such that

$$d_1 = v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2})). \quad (2.5)$$

Moreover, $(0, \theta_{d_2, m_2, q_2})$ is linearly stable when $d_2^* < d_2 < v_1(e^{\alpha_2 A_2} m_2)$, and unstable for all $d_2 < d_2^*$.

Theorem 2.3 demonstrates that nonlinear boundaries can create intermediate stability regimes where linear boundaries would typically yield monotonic stability behavior. Theorems 2.2 and 2.3 means that the quadratic boundary terms modify the principal eigenvalue conditions governing stability transitions and introduce non-monotonic dependence of stability on diffusion rates through their influence on θ profiles. In addition, the stability criteria depend explicitly on weighted spatial averages involving $e^{\alpha_i A_i}$, showing how boundary effects couple with environmental heterogeneity $A_i(x)$. Theorems 2.2 and 2.3 have clear biological implications. Namely, density-dependent dispersal at boundaries can prevent predator invasion at intermediate mobility ranges, create new coexistence regimes near habitat edges, and induce threshold effects in species establishment. Moreover, the stability thresholds d_2^0 and d_2^* represent critical mobility thresholds for species persistence.

The above two theorems mean that there may exist stability switches for the two semi-trivial steady-state solutions $(\theta_{d_1, m_1, q_1}, 0)$ and $(0, \theta_{d_2, m_2, q_2})$ when the diffusion coefficients d_1 and d_2 vary. This stability change indicates potential bifurcation points where positive coexistence states may emerge. Denote the two semi-trivial steady-state solution branches with parameter d_2 by

$$\begin{aligned}\Gamma_u &= \{(d_2, \theta_{d_1, m_1, q_1}, 0) : (d_1, d_2) \in \mathbb{D}_1 \times \mathbb{R}_+\}, \\ \Gamma_v &= \{(d_2, 0, \theta_{d_2, m_2, q_2}) : (d_1, d_2) \in \mathbb{R}_+ \times \mathbb{D}_2\}.\end{aligned}$$

Our goal is to look for positive steady states emanating from the branches Γ_u and Γ_v . For this purpose, we first consider the nonexistence and boundedness of positive steady-state solutions.

Theorem 2.4 (Nonexistence and A Priori Bounds). *Under conditions (A1) and (A2), system (2.1) admits no positive steady-state solution when $d_1 \geq v_1(e^{\alpha_1 A_1} m_1)$. For $d_1 < v_1(e^{\alpha_1 A_1} m_1)$, every positive steady-state solution (u, v) of (2.1) satisfies the uniform estimates:*

$$\|u\|_{C(\bar{\Omega})} < \bar{u}, \quad \|v\|_{C(\bar{\Omega})} < \bar{v},$$

where $\|w\|_{C(\bar{\Omega})} := \max_{\bar{\Omega}} w$ and

$$\bar{u} := e^{\alpha_1 \|A_1\|_{C(\bar{\Omega})}} \|m_1\|_{C(\bar{\Omega})}, \quad \bar{v} := e^{\alpha_2 \|A_2\|_{C(\bar{\Omega})}} \left[\|m_2\|_{C(\bar{\Omega})} + b e^{\alpha_1 \|A_1\|_{C(\bar{\Omega})}} \bar{u} \right]. \quad (2.6)$$

Denote

$$\mathbb{D}_{01} := \begin{cases} \mathbb{R}_+ & \text{if } \int_{\Omega} e^{\alpha_1 A_1} m_1 dx \geq 0, \\ \mathbb{D}_{11} & \text{if } \int_{\Omega} e^{\alpha_1 A_1} m_1 dx < 0, \end{cases}$$

where

$$\mathbb{D}_{11} := (0, v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \bar{v})) + \delta) \cup (v_1(e^{\alpha_1 A_1} m_1) - \delta, v_1(e^{\alpha_1 A_1} m_1))$$

for some sufficiently small $\delta > 0$ and in which $\bar{v}$ is as defined in (2.6). Then the global bifurcation structure is as follows.

Theorem 2.5 (Global Bifurcation Structure). *Under conditions (A1), (A2), (H1), and (2.2), the set of positive steady-state solutions of (2.1) contains a continuum S_{01} in $\mathbb{R}_+ \times \mathbb{X}^2$ (where $\mathbb{X} = W^{2,p}(\bar{\Omega})$) that bifurcates from the semi-trivial solution branch Γ_u at $(d_2^0, \theta_{d_1, m_1, q_1}, 0)$.*

- (i) *For $d_1 \in \mathbb{D}_{01}$, the projection of S_{01} on d_2 -axis: $\text{Proj}_{d_2} S_{01} \supset (d_2^0, +\infty)$ and so (2.1) has at least one positive steady-state solution for all $d_2 > d_2^0$.*
- (ii) *For $d_1 \in \mathbb{D}_1 \setminus \mathbb{D}_{01}$, S_{01} forms a bounded continuum connecting to $(0, \theta_{d_2, m_2, q_2})$ at $d_2 = d_2^*$ (uniquely determined by the implicit condition in (2.5)), and so (2.1) has at least one positive steady-state solution for d_2 lying between d_2^* and d_2^0 .*

This theorem reveals a fundamental dichotomy in predator-prey dynamics governed by habitat quality through the prey diffusion parameter d_1 . In favorable habitats ($d_1 \in \mathbb{D}_{01}$), system (2.1) exhibits robust coexistence for all predator mobility levels beyond the critical threshold d_2^0 , where this bifurcation point represents the minimum dispersal capability predators require to successfully invade an established prey population. The persistence of coexistence regardless of increasing d_2 suggests that high-quality habitats can buffer against predator over-dispersion effects.

More nuanced dynamics emerge in marginal habitats ($d_1 \in \mathbb{D}_1 \setminus \mathbb{D}_{01}$), where coexistence becomes constrained to an intermediate predator mobility window (d_2^*, d_2^0) . This finite parameter range reflects competing ecological constraints: at low mobility ($d_2 < d_2^*$), predators cannot effectively track prey distributions, while excessive dispersal ($d_2 > d_2^0$) leads to population unsustainability through over-dispersion. The existence of this optimal mobility window is mediated by the nonlinear boundary terms $q_1 u^2$ and $q_2 v^2$, which create density-dependent barriers at habitat edges that fundamentally alter system dynamics compared to classical linear boundary models. These boundary effects generate three key ecological phenomena: density-dependent regulation of population fluxes at habitat edges, non-monotonic stability properties that depend sensitively on dispersal rates, and the emergence of intermediate optimal mobility ranges that explain observed edge-associated species distributions in fragmented landscapes. Theorem 2.5 thus provides both a rigorous analytical framework for studying nonlinear boundary value problems and testable biological predictions about how habitat edges mediate coexistence through dispersal constraints.

The mathematical significance of Theorem 2.5 lies in establishing the first complete global bifurcation analysis for this class of nonlinear boundary problems, while the ecological value stems from quantifying how habitat quality (via d_1) interacts with predator mobility (d_2) to determine coexistence patterns through boundary-mediated effects. These results offer mechanistic explanations for field observations of mobility-dependent species distributions while providing new mathematical tools for studying edge effects in ecological systems.

Theorem 2.6 (Global Bifurcation Structure). *In addition to conditions (A1), (A2), (H2), and (2.4), assume that θ_{d_2, m_2, q_2} is locally monotonic near $d_2 = d_2^*$ and*

$$\int_{\Omega} e^{\alpha_1 A_1} m_1 dx < 0. \quad (2.7)$$

Then the set of positive steady-state solutions of (2.1) contains a continuum S_{02} in $\mathbb{R}_+ \times \mathbb{X}^2$, emanating from the solution branch Γ_v at $(d_2^, 0, \theta_{d_2, m_2, q_2})$. Furthermore, when $d_1 \in (v_1(e^{\alpha_1 A_1} (m_1 - a m_2^+)), v_1(e^{\alpha_1 A_1} m_1))$, there exists a unique $d_2^\infty \neq d_2^*$ such that $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2}) \in S_{02}$, where d_2^∞ is uniquely determined by $d_1 = v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$. Consequently, system (2.1) admits at least one positive steady-state solution for d_2 lying between d_2^∞ and d_2^* , where the relative ordering of d_2^∞ and d_2^* remains indeterminate.*

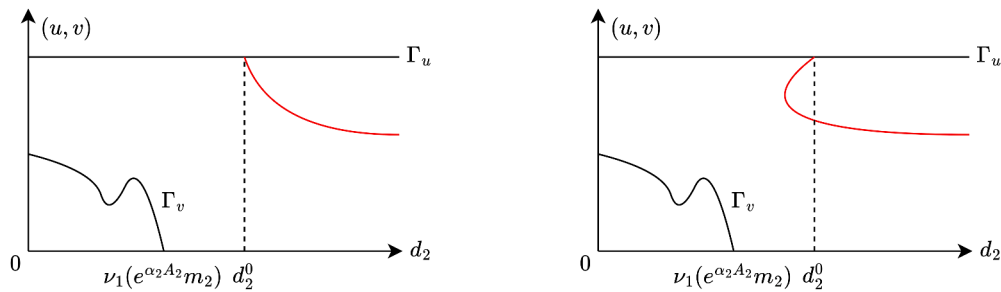


Fig. 1. Possible bifurcation diagram of coexistence states in the case $d_1 \in \mathbb{D}_{01}$.

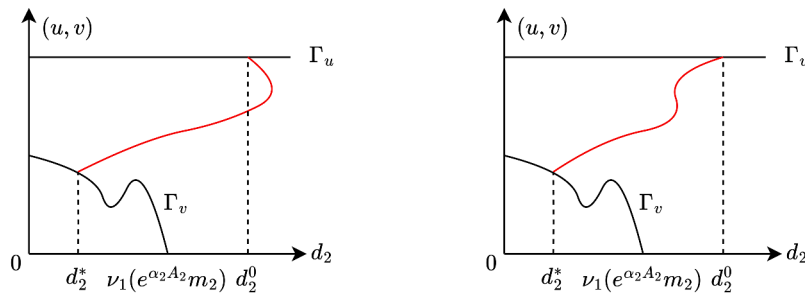


Fig. 2. Possible bifurcation diagram of coexistence states in the case $d_1 \in \mathbb{D}_1 \setminus \mathbb{D}_{01}$.

Theorem 2.6 establishes a sophisticated bifurcation structure emerging from the predator-only solution branch Γ_v , revealing how nonlinear boundary conditions and habitat heterogeneity interact to shape coexistence patterns. Under the specified conditions, including the critical requirement that the spatial average of weighted prey growth rates is negative (i.e., (2.7)), system (2.1) exhibits a continuum S_{02} of positive steady states bifurcating from Γ_v at the critical predator diffusion rate d_2^* . The requirement (2.7) is crucial. Indeed, when (2.7) doesn't hold, Lemma 3.2 implies that $v_1(e^{\alpha_1 A_1} m_1) < 0$, which would render the interval $d_1 \in (v_1(e^{\alpha_1 A_1} (m_1 - am_2^+)), v_1(e^{\alpha_1 A_1} m_1))$ vacuous for positive diffusion rates $d_1 > 0$. The local monotonicity assumption of θ_{d_2, m_2, q_2} near d_2^* plays a pivotal role, ensuring the existence of a unique secondary connection point d_2^∞ where the solution continuum S_{02} closes back to the predator-only branch. This bifurcation structure creates a parameter region (between d_2^* and d_2^∞) where stable coexistence occurs, with the relative ordering of these critical values depending on subtle interactions between the species' diffusion rates and environmental gradients.

Biologically, these mathematical results translate into important ecological insights. The negative spatial average condition suggests prey face significant growth limitations in parts of the habitat, while the local monotonicity reflects how predator populations respond smoothly to dispersal changes near the critical threshold. The emergence of the d_2^∞ connection point indicates that predator-prey coexistence becomes constrained within an intermediate mobility window: too little movement prevents effective prey tracking, while excessive dispersal leads to population dilution. The most striking implication of Theorem 2.6 is the indeterminate ordering of d_2^* and d_2^∞ , which mathematically stems from the nonlinear boundary terms creating multiple possible stability configurations. Ecologically, this means the same habitat can exhibit different coexistence patterns depending on initial conditions and historical contingencies, potentially explaining field observations of alternative stable states in edge-dominated ecosystems.

These results significantly advance our understanding of nonlinear boundary effects by providing the first complete characterization of solution continua connecting to predator-only states, quantifying how habitat quality (via m_1 distribution) mediates edge effects on coexistence, and revealing how nonlinear dispersal responses at boundaries can generate multiple stability regimes. The analysis offers both rigorous mathematical tools for studying nonlinear boundary value problems and testable predictions about how edge-mediated dispersal regulates species coexistence in fragmented landscapes.

The global bifurcation structures established in Theorems 2.5 and 2.6 are visually represented in Figs. 1–3, providing crucial insights into the solution landscape of our system. Figs. 1 and 2 illustrate the complete bifurcation scenario described by Theorem 2.5, demonstrating both the unbounded solution branch for favorable habitat conditions ($d_1 \in \mathbb{D}_{01}$) and the bounded continuum emerging under marginal habitat conditions ($d_1 \in \mathbb{D}_1 \setminus \mathbb{D}_{01}$). Fig. 3 specifically captures the intricate connection structure predicted by Theorem 2.6, where the solution branch reconnects to the predator-only equilibrium at a secondary bifurcation point. To complement our theoretical analysis, we present a carefully designed numerical experiment that validates these mathematical predictions. The simulation employs a biologically meaningful piecewise-defined growth rate function:

$$m(x) = \begin{cases} 1, & x \leq 4.5, \\ 1 - 18(x - 4.5), & x > 4.5. \end{cases}$$

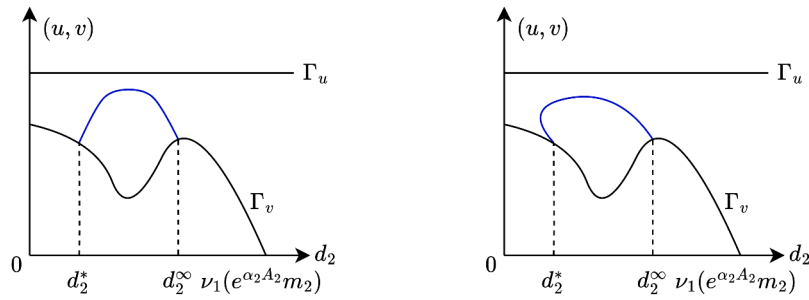
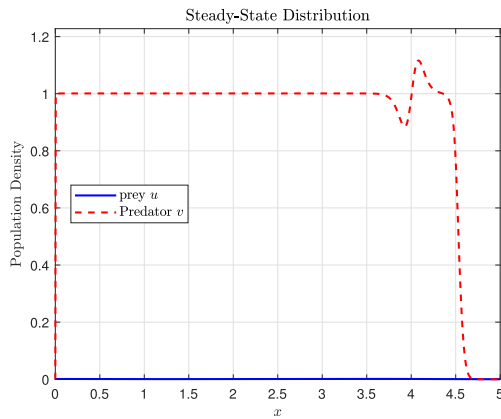
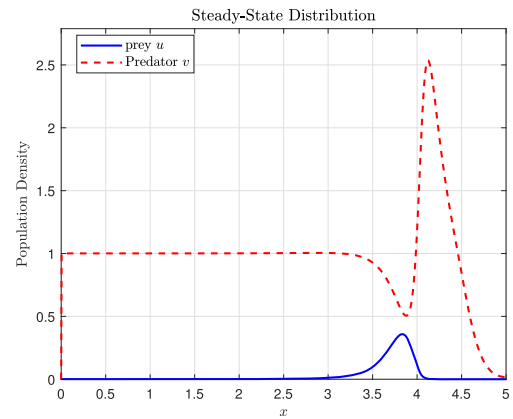


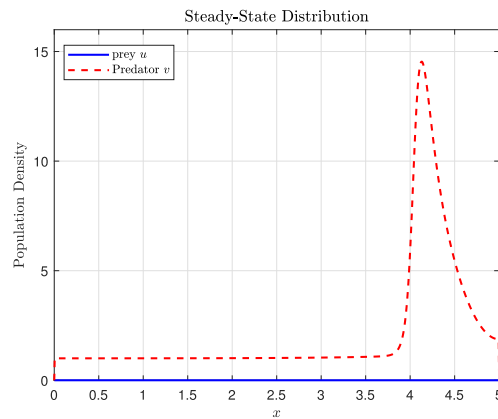
Fig. 3. Possible bifurcation diagram of coexistence states in the case $d_1 \in (\nu_1(e^{\alpha_1 A_1}(m_1 - am_2^+)), \nu_1(e^{\alpha_1 A_1}m_1))$ and $d_2^* < d_2^\infty$.



(a) $d_2 = 0.001$



(b) $d_2 = 0.05$



(c) $d_2 = 0.6$

Fig. 4. Solution $(u(x, 500), v(x, 500))$ of (1.4) in the interval $\Omega = (0, 5)$, where the initial value are: $u_0(x) = 0.1 + 0.1 \sin(5x)$, $v_0(x) = 0.1 + 0.1 \sin(5x)$, and parameter values are: $d_1 = 0.001$, $\alpha_1 = \alpha_2 = 1$, $a = b = 1$, $q_1 = q_2 = -1$, $A_1(x) = A_2(x) = \frac{2}{1+e^{-20(x-4)}} - 1$, $m_1(x) = m_2(x) = m(x)$.

This function models a habitat with an abrupt transition from favorable to unfavorable conditions at $x = 4.5$, creating a strong environmental gradient that tests our theoretical predictions under extreme conditions. The resulting numerical solutions, displayed in Fig. 4, exhibit excellent agreement with our analytical results while revealing additional insights into the spatial distribution patterns of species near habitat boundaries.

The numerical analysis serves three critical roles: verifying theoretical bifurcation thresholds, confirming the stability of solutions under non-smooth environmental shifts, and visualizing the interplay between nonlinear boundary effects and habitat gradients. This combination of rigorous analysis and numerical validation strengthens the biological relevance of our mathematical framework while maintaining the highest standards of theoretical integrity.

3. A single-species model

In this section, we consider the following single-species model with nonlinear boundary condition:

$$\begin{cases} \frac{\partial u}{\partial t} = d \nabla \cdot [\nabla u - \alpha u \nabla A(x)] + u[m(x) - u] & \text{in } \Omega \times \mathbb{R}_+, \\ d \left[\frac{\partial u}{\partial n} - \alpha u \frac{\partial A(x)}{\partial n} \right] = q(x)u^2 & \text{on } \partial\Omega \times \mathbb{R}_+, \end{cases} \quad (3.1)$$

where $A \in C^{2+\sigma}(\overline{\Omega})$, $m \in C^\sigma(\overline{\Omega})$ satisfies $\int_\Omega m dx > 0$ and $\min_\Omega m < 0$, $q \in C^\sigma(\overline{\partial\Omega})$ for some $\sigma \in (0, 1)$ and satisfies $q(x) \leq 0$ for all $x \in \partial\Omega$. Let $w = e^{-\alpha A}u$, then we can obtain the following system equivalent to (3.1):

$$\begin{cases} e^{\alpha A} \frac{\partial w}{\partial t} = d \nabla \cdot [e^{\alpha A} \nabla w] + e^{\alpha A} w[m(x) - e^{\alpha A} w] & \text{in } \Omega, \\ d \frac{\partial w}{\partial n} = q(x)e^{\alpha A} w^2 & \text{on } \partial\Omega. \end{cases} \quad (3.2)$$

By linearizing (3.2) at $w = 0$, we have the following linear eigenvalue problem:

$$\begin{cases} \mu e^{\alpha A} w = d \nabla \cdot [e^{\alpha A} \nabla w] + e^{\alpha A} m(x)w & \text{in } \Omega, \\ \frac{\partial w}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.3)$$

In order to investigate the above eigenvalue problem, we consider the first eigenvalue $\mu_1(d, m, h)$ of the following eigenvalue problem

$$\begin{cases} \mu e^{\alpha A} w = d \nabla \cdot [e^{\alpha A} \nabla w] + e^{\alpha A} m(x)w & \text{in } \Omega, \\ d \frac{\partial w}{\partial n} = h(x)w & \text{on } \partial\Omega, \end{cases} \quad (3.4)$$

where $h(x) \in C^\sigma(\overline{\partial\Omega})$ for some $\sigma \in (0, 1)$. Obviously, the variational characterization of $\mu_1(d, m, h)$ of (3.4) is as follows:

$$\mu_1(d, m, h) = \sup_{\substack{w \in W^{1,2}(\Omega) \\ w \neq 0}} \left\{ \frac{-d \int_\Omega e^{\alpha A} |\nabla w|^2 dx + \int_{\partial\Omega} e^{\alpha A} h w^2 ds + \int_\Omega m e^{\alpha A} w^2 dx}{\int_\Omega e^{\alpha A} w^2 dx} \right\}. \quad (3.5)$$

Thus, $\mu_1(d, m, h)$ is increasing with respect to m and h .

Obviously, the variational characterization of the first eigenvalue of (3.3) is $\mu_1(d, m, 0)$. In particular, $\mu_1(d, m, 0) = 0$ if and only if $\lambda = \frac{1}{d}$ is a principal eigenvalue of the following eigenvalue problem with indefinite weight

$$\begin{cases} -\nabla \cdot [e^{\alpha A} \nabla w] = \lambda e^{\alpha A} m(x)w & \text{in } \Omega, \\ \frac{\partial w}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.6)$$

We say that λ is a principal eigenvalue if (3.6) has a positive solution. (Notice that 0 is always a principal eigenvalue.) Regarding the existence of such a principal eigenvalue, we recall the following result from [65,66].

Lemma 3.1. Assume that $m(x)$ is positive on an open subset of Ω and $\int_\Omega e^{\alpha A} m dx < 0$. Then problem (3.6) admits a positive principal eigenvalue $\sigma_1(e^{\alpha A} m)$ characterized variationally by:

$$\frac{1}{\sigma_1(e^{\alpha A} m)} = v_1(e^{\alpha A} m) \triangleq \sup_{\substack{w \in W^{1,2}(\Omega) \\ w \neq 0}} \left\{ \frac{\int_\Omega m e^{\alpha A} w^2 dx}{\int_\Omega e^{\alpha A} |\nabla w|^2 dx} \right\}. \quad (3.7)$$

The principal eigenvalue is the only positive eigenvalue possessing a strictly positive eigenfunction, and the corresponding eigenspace is one-dimensional. The principal eigenvalue depends continuously in m with respect to $L^p(\Omega)$ for any $p \in (1, +\infty]$ in the case of one or two space dimensions, and for $p > N/2$ in the case of $N > 2$ space dimensions. Moreover, the positive principal eigenvalue $\sigma_1(e^{\alpha A} m)$ is decreasing with respect to m in the sense that if $m_1 \geq m_2$ then $\sigma_1(e^{\alpha A} m_1) \leq \sigma_1(e^{\alpha A} m_2)$, and if $m_1 > m_2$ on a subset of positive measure then $\sigma_1(e^{\alpha A} m_1) < \sigma_1(e^{\alpha A} m_2)$.

Lemma 3.2. Assume that $m(x)$ is negative on an open subset of Ω and $\int_\Omega e^{\alpha A} m dx \geq 0$. Then the problem (3.6) admits a negative principal eigenvalue $\sigma_1(e^{\alpha A} m)$ determined by (3.7).

For the existence of principal eigenvalues in weighted eigenvalue problems, we refer to the classical works [67,68]. A comprehensive abstract theory that encompasses general boundary operators is developed in Chapter 9 of López-Gómez [69], which also provides a systematic summary of the fundamental properties of these eigenvalues. In view of the above lemmas, along with Theorem 2.2.1 and Corollary 2.2.2 in [66], we obtain the following results.

Lemma 3.3.

- (i) Under assumptions of Lemma 3.2, $\mu_1(d, m, 0) > 0$ for all $d > 0$.
- (ii) Under assumptions of Lemma 3.1, $\mu_1(d, m, 0) > 0$ for all $0 < d < v_1(e^{\alpha A} m)$, $\mu_1(d, m, 0) = 0$ if $d = v_1(e^{\alpha A} m)$, and $\mu_1(d, m, 0) < 0$ for all $d > v_1(e^{\alpha A} m)$.

- (iii) $v_1(e^{\alpha A} m)$ is increasing with respect to m in the sense that if $m_1 \geq m_2$ then $v_1(e^{\alpha A} m_1) \geq v_1(e^{\alpha A} m_2)$, and if $m_1 > m_2$ on a subset of positive measure then $v_1(e^{\alpha A} m_1) > v_1(e^{\alpha A} m_2)$.

Similar to Lemma 4.2 and Lemma 4.4 in [70] (see also Lemma 2.6 and Lemma 2.7 in [51]), we have the existence, uniqueness and its properties of positive steady-state solutions of (3.2).

Lemma 3.4. System (3.2) has a unique positive steady-state solution $\theta_{d,m,q}$ provided that

$$\text{either } \int_{\Omega} e^{\alpha A} m \, dx \geq 0, \text{ or } \int_{\Omega} e^{\alpha A} m \, dx < 0 \text{ and } 0 < d < v_1(e^{\alpha A} m). \quad (3.8)$$

Furthermore, the following hold:

- (i) $\int_{\Omega} e^{\alpha A} m \, dx + \int_{\partial\Omega} q(x) e^{2\alpha A} \theta_{d,m,q} \, ds < \int_{\Omega} e^{2\alpha A} \theta_{d,m,q} \, dx$.
- (ii) $d \mapsto \theta_{d,m,q}$ is continuous from $\mathbb{D}$ to $W^{2,p}(\Omega) \cap C^2(\overline{\Omega})$ and satisfies $e^{\alpha A} \theta_{d,m,q} \rightarrow m^+$ as $d \rightarrow 0^+$ where $m^+(x) = \max\{m(x), 0\}$,

$$\mathbb{D} = \begin{cases} \mathbb{R}_+ & \text{if } \int_{\Omega} e^{\alpha A} m \, dx \geq 0, \\ (0, v_1(e^{\alpha A} m)) & \text{if } \int_{\Omega} e^{\alpha A} m \, dx < 0. \end{cases}$$
- (iii) If $\int_{\Omega} e^{\alpha A} m \, dx \geq 0$ then $\theta_{d,m,q} \rightarrow \frac{\int_{\Omega} e^{\alpha A} m \, dx}{\int_{\Omega} e^{2\alpha A} \, dx - \int_{\partial\Omega} q e^{2\alpha A} \, ds}$ as $d \rightarrow \infty$.
- (iv) If $\int_{\Omega} e^{\alpha A} m \, dx < 0$ then $\theta_{d,m,q} \rightarrow 0$ as $d \rightarrow v_1(e^{\alpha A} m) - 0$.
- (v) Assume that $w \in C^{\sigma}(\overline{\Omega})$ with $\sigma \in (0, 1)$ is nonconstant such that $m \leq (\neq) w$ and (3.8) holds for m replaced by w , then $\theta_{d,m,q} < \theta_{d,w,q}$.
- (vi) Assume that $h \in C^{\sigma}(\overline{\partial\Omega})$ with $\sigma \in (0, 1)$ and satisfies $h \leq (\neq) q \leq 0$ on $\partial\Omega$, then $\theta_{d,m,h} < \theta_{d,m,q}$.
- (vii) $\inf_{\Omega} \theta_{d,m,q} \geq \inf_{\Omega} e^{-\alpha A} m$ and $\sup_{\Omega} \theta_{d,m,q} \leq \sup_{\Omega} e^{-\alpha A} m$.

4. Semi-trivial steady states

In view of Lemma 3.4, we have the existence of semi-trivial steady states as follows.

Lemma 4.1. Under conditions (A1) and (A2), system (2.1) admits a semi-trivial steady-state solution $(u, v) = (\theta_{d_1, m_1, q_1}, 0)$ provided that (H1) hold, and a semi-trivial steady-state solution $(u, v) = (0, \theta_{d_2, m_2, q_2})$ provided that (H2) hold.

We next discuss the stability of semi-trivial steady states. Linearizing the corresponding elliptic system of (2.1) at a positive steady state $\theta = (\theta_1, \theta_2)$, we have

$$\begin{cases} \mu u = d_1 e^{-\alpha_1 A_1} \nabla \cdot [e^{\alpha_1 A_1} \nabla u] \\ \quad + u[m_1 - 2e^{\alpha_1 A_1} \theta_1 - ae^{\alpha_2 A_2} \theta_2] - ae^{\alpha_2 A_2} \theta_1 v & \text{in } \Omega, \\ \mu v = d_2 e^{-\alpha_2 A_2} \nabla \cdot [e^{\alpha_2 A_2} \nabla v] \\ \quad + v[m_2 - 2e^{\alpha_2 A_2} \theta_2 + be^{\alpha_1 A_1} \theta_1] + be^{\alpha_1 A_1} \theta_2 u & \text{in } \Omega, \\ d_1 \frac{\partial u}{\partial n} = 2q_1 e^{\alpha_1 A_1} \theta_1 u & \text{on } \partial\Omega, \\ d_2 \frac{\partial v}{\partial n} = 2q_2 e^{\alpha_2 A_2} \theta_2 v & \text{on } \partial\Omega. \end{cases} \quad (4.1)$$

μ_1 is said to be the first eigenvalue of (4.1) if it is an eigenvalue with the greatest real part among all eigenvalues of the linearized eigenvalue problem (4.1). In view of [71], we have the following result.

Lemma 4.2. Let μ_1 be the first eigenvalue of linearized eigenvalue problem (4.1) associated the positive steady state $\theta = (\theta_1, \theta_2)$ of (2.1). Then the positive steady state θ of (2.1) is linearly stable (respectively, linearly unstable) if $\mu_1 < 0$ (respectively, $\mu_1 > 0$).

It follows from Lemma 4.2 that the stability of the semi-trivial steady-state solution $(\theta_{d_1, m_1, q_1}, 0)$ is determined by the first eigenvalue $\mu_1(d_2, m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}, 0)$ of the eigenvalue problem

$$\begin{cases} \mu v = d_2 e^{-\alpha_2 A_2} \nabla \cdot [e^{\alpha_2 A_2} \nabla v] + v[m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}] & \text{in } \Omega, \\ \frac{\partial v}{\partial n} = 0 & \text{on } \partial\Omega, \end{cases} \quad (4.2)$$

while the stability of the semi-trivial steady-state solution $(0, \theta_{d_2, m_2, q_2})$ is determined by the first eigenvalue $\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 0)$ of the eigenvalue problem

$$\begin{cases} \mu u = d_1 e^{-\alpha_1 A_1} \nabla \cdot [e^{\alpha_1 A_1} \nabla u] + u[m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}] & \text{in } \Omega, \\ \frac{\partial u}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases} \quad (4.3)$$

Thus, we have the following criterion.

Lemma 4.3. Under conditions (A1), (A2), (H1), and (H2), the linear stability of semi-trivial steady-state solution $(\theta_{d_1, m_1, q_1}, 0)$ and $(0, \theta_{d_2, m_2, q_2})$ is determined by the sign of $\mu_1(d_2, m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}, 0)$ and $\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 0)$, respectively. In particular, $(0, 0)$ is always linearly unstable.

In view of Lemmas 3.3, 3.4 and 4.3, we have the following results.

Lemma 4.4.

(i) Under conditions (A1), (A2), and (H1), the semi-trivial steady-state solution $(\theta_{d_1, m_1, q_1}, 0)$ is unstable provided that either

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx \geq 0, \quad \min_{\Omega} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) < 0 \quad \text{and} \quad d_2 > 0$$

or

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx < 0 \quad \text{and} \quad 0 < d_2 < v_1(e^{\alpha_2 A_2} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1})),$$

and is asymptotically stable provided that

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx < 0 \quad \text{and} \quad d_2 > v_1(e^{\alpha_2 A_2} (m_2 + be^{\alpha_1 A_1} \theta_{d_1, m_1, q_1})).$$

(ii) Under conditions (A1), (A2), and (H2), the semi-trivial steady-state solution $(0, \theta_{d_2, m_2, q_2})$ is unstable provided that either

$$\int_{\Omega} e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) dx \geq 0 \quad \text{and} \quad d_1 > 0$$

or $\max_{\Omega} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) > 0$,

$$\int_{\Omega} e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) dx < 0 \quad \text{and} \quad 0 < d_1 < v_1(e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2})),$$

and is asymptotically stable provided that $\max_{\Omega} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) > 0$,

$$\int_{\Omega} e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) dx < 0 \quad \text{and} \quad d_1 > v_1(e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2})).$$

(iii) Under conditions (A1), (A2), (H2), and (2.4), there exists $d_2^* < v_1(e^{\alpha_2 A_2} m_2)$ depending on d_1 and satisfying (2.5) such that $(0, \theta_{d_2, m_2, q_2})$ is linearly stable for $d_2^* < d_2 < v_1(e^{\alpha_2 A_2} m_2)$ and is unstable for $d_2 < d_2^*$.

Proof. Conclusions (i) and (ii) follow directly from Lemmas 3.3 and 4.3. It follows from Lemma 3.4 that $\theta_{d_2, m_2, q_2} \rightarrow 0$ as $d_2 \rightarrow v_1(e^{\alpha_2 A_2} m_2) - 0$ and hence

$$\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 0) \rightarrow \mu_1(d_1, m_1, 0) \quad \text{as} \quad d_2 \rightarrow v_1(e^{\alpha_2 A_2} m_2) - 0.$$

Notice that

$$e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2} \rightarrow m_2^+ \quad \text{as} \quad d_2 \rightarrow 0,$$

then we have

$$\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 0) \rightarrow \mu_1(d_1, m_1 - am_2^+, 0) \quad \text{as} \quad d_2 \rightarrow 0.$$

Moreover, $\mu_1(d_1, m_1, 0) > 0$ and $\mu_1(d_1, m_1 - am_2^+, 0) < 0$ for $v_1(e^{\alpha_1 A_1} m_1) > d_1 > v_1(e^{\alpha_1 A_1} (m_1 - am_2^+))$ by Lemma 3.3. Thus, there exists $d_2^* < v_1(e^{\alpha_2 A_2} m_2)$ such that

$$\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}, 0) = 0,$$

that is, $d_1 = v_1(e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}))$. Moreover, $\mu_1(d_1, m_1 - ae^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 0)$ is negative for $v_1(e^{\alpha_2 A_2} m_2) > d_2 > d_2^*$ and is positive for $d_2 < d_2^*$. The proof is completed. $\square$

5. A priori estimates

In this section, we will show some a priori estimates for steady-state solution to (2.1), which is equivalent to

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u] = e^{\alpha_1 A_1} u [m_1(x) - e^{\alpha_1 A_1} u - ae^{\alpha_2 A_2} v] & \text{in } \Omega, \\ -d_2 \nabla \cdot [e^{\alpha_2 A_2} \nabla v] = e^{\alpha_2 A_2} v [m_2(x) - e^{\alpha_2 A_2} v + be^{\alpha_1 A_1} u] & \text{in } \Omega, \\ d_1 \frac{\partial u}{\partial n} = q_1(x) e^{\alpha_1 A_1} u^2 & \text{on } \partial\Omega, \\ d_2 \frac{\partial v}{\partial n} = q_2(x) e^{\alpha_2 A_2} v^2 & \text{on } \partial\Omega. \end{cases} \quad (5.1)$$

It is well known that the priori estimates are the fundamental in yielding the existence and nonexistence of positive solution of (5.1). We first provide a simple but useful proposition by virtue of the Maximum Principle, before proving the $C(\bar{\Omega})$ -estimate for positive solutions of (5.1).

Proposition 5.1. Suppose that $g \in C(\overline{\Omega} \times \mathbb{R})$.

(i) If $w \in C^2(\Omega) \cap C^1(\overline{\Omega})$ satisfies

$$\begin{cases} \nabla \cdot [e^{\alpha A} \nabla w] + g(x, w(x)) \geq 0 & \text{in } \Omega, \\ \frac{\partial w}{\partial n} \leq 0 & \text{on } \partial\Omega, \end{cases}$$

and $w(x_0) = \max_{\overline{\Omega}} w$, then $g(x_0, w(x_0)) \geq 0$.

(ii) If $w \in C^2(\Omega) \cap C^1(\overline{\Omega})$ satisfies

$$\begin{cases} \nabla \cdot [e^{\alpha A} \nabla w] + g(x, w(x)) \leq 0 & \text{in } \Omega, \\ \frac{\partial w}{\partial n} \geq 0 & \text{on } \partial\Omega, \end{cases}$$

and $w(x_0) = \min_{\overline{\Omega}} w$, then $g(x_0, w(x_0)) \leq 0$.

Proof. The proof of this Proposition is very similar to that of Proposition 2.2 in [72], so it is omitted here. $\square$

We now prove the $C(\overline{\Omega})$ -estimate for positive solutions of (5.1).

Proof of Theorem 2.4. Let (u, v) be a positive solution of (5.1). Multiplying the equation about u in (5.1) by u and integrating the result by parts, we have

$$d_1 \int_{\Omega} e^{\alpha_1 A_1} |\nabla u|^2 dx = \int_{\Omega} e^{\alpha_1 A_1} u^2 [m_1 - e^{\alpha_1 A_1} u - a e^{\alpha_2 A_2} v] dx + \int_{\partial\Omega} q_1 e^{\alpha_1 A_1} u^3 ds.$$

It follows from $(u, v) > 0$ and $q_1 \leq 0$ that

$$v_1(e^{\alpha_1 A_1} m_1) \geq \frac{\int_{\Omega} m_1 e^{\alpha_1 A_1} u^2 dx}{\int_{\Omega} e^{\alpha_1 A_1} |\nabla u|^2 dx} > d_1.$$

Hence, (5.1) has no positive solution when $d_1 \geq v_1(e^{\alpha_1 A_1} m_1)$.

Now we assume that $d_1 < v_1(e^{\alpha_1 A_1} m_1)$, and (u, v) is a positive solution of (5.1). For the equations of u ,

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u] = e^{\alpha_1 A_1} u [m_1 - e^{\alpha_1 A_1} u - a e^{\alpha_2 A_2} v] & \text{in } \Omega, \\ d_1 \frac{\partial u}{\partial n} = q_1 e^{\alpha_1 A_1} u^2 & \text{on } \partial\Omega. \end{cases} \quad (5.2)$$

Note that $q_1 \leq 0$, then by Proposition 5.1, we have

$$e^{\alpha_1 A_1} u [m_1 - e^{\alpha_1 A_1} u - a e^{\alpha_2 A_2} v] \geq 0$$

at x_0 , where $u(x_0) = \|u\|_{C(\overline{\Omega})}$. It follows from $(u, v) > 0$ that

$$\begin{aligned} u(x_0) &\leq e^{-\alpha_1 A_1(x_0)} [m_1(x_0) - a e^{\alpha_2 A_2(x_0)} v(x_0)] \\ &< e^{-\alpha_1 A_1(x_0)} m_1(x_0) \\ &\leq e^{\alpha_1 \|A_1\|_{C(\overline{\Omega})}} \|m_1\|_{C(\overline{\Omega})}. \end{aligned}$$

Similarly, for the equations v , by $q_2 \leq 0$ and Proposition 5.1, we have

$$\begin{aligned} v(x_1) &\leq e^{-\alpha_2 A_2(x_1)} [m_2(x_1) + b e^{\alpha_1 A_1(x_1)} u(x_1)] \\ &< e^{\alpha_2 \|A_2\|_{C(\overline{\Omega})}} \left[\|m_2\|_{C(\overline{\Omega})} + b e^{\alpha_1 \|A_1\|_{C(\overline{\Omega})}} \bar{u} \right], \end{aligned}$$

where $v(x_1) = \|v\|_{C(\overline{\Omega})}$ and $\bar{u}$ is as defined in (2.6). Moreover, the $C(\overline{\Omega})$ -estimate for positive steady-state solutions (u_1, u_2) of (1.4) follows from $u = e^{-\alpha_1 A_1} u_1$ and $v = e^{-\alpha_2 A_2} u_2$. This ends the proof. $\square$

6. Local bifurcation

This section use the well-known bifurcation results from Crandall and Rabinowitz [45] to study the coexisting steady-state bifurcations arising from the semi-trivial solutions of (5.1) by regarding d_2 as a parameter, under the conditions (A1), (A2), (H1) and (H2).

Recall that the two semi-trivial solution branches with parameter d_2 are Γ_u and Γ_v .

Clearly, the conditions (A1), (A2) and (H1) ensure the existence of the semi-trivial solution branch Γ_u , while (A1), (A2) and (H2) ensure the existence of branch Γ_v . We first analyze the steady-state bifurcation from the branch Γ_u at $d_2 = d_2^0 \triangleq v_1(e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}))$ under the condition (2.2), that is,

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) dx < 0.$$

Define the associated operator $\mathcal{O} : \mathbb{R}_+ \times \mathbb{X}^2 \mapsto \mathbb{Y}^2$ by

$$\mathcal{O}(d_2, u, v) = \begin{bmatrix} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u] - e^{\alpha_1 A_1} u [m_1 - e^{\alpha_1 A_1} u - a e^{\alpha_2 A_2} v] \\ -d_2 \nabla \cdot [e^{\alpha_2 A_2} \nabla v] - e^{\alpha_2 A_2} v [m_2 - e^{\alpha_2 A_2} v + b e^{\alpha_1 A_1} u] \\ d_1 \frac{\partial u}{\partial \mathbf{n}} - q_1 e^{\alpha_1 A_1} u^2 \\ d_2 \frac{\partial v}{\partial \mathbf{n}} - q_2 e^{\alpha_2 A_2} v^2 \end{bmatrix}, \quad (6.1)$$

where $\mathbb{X} = W^{2,p}(\overline{\Omega})$ and $\mathbb{Y} = L^p(\Omega) \times W^{(p-1)/p,p}(\partial\Omega)$. It is not difficult to verify that $(u, v) \in \mathbb{X}^2$ is a nonnegative solution of system (5.1) if and only if $\mathcal{O}(d_2, u, v) = 0$. By straightforward calculations, the Fréchet derivative of $\mathcal{O}$ with respect to (u, v) at $(d_2, u, v) = (d_2^0, \theta_{d_1, m_1, q_1}, 0)$ is given by

$$\mathcal{L}(u, v) = \begin{bmatrix} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u] - e^{\alpha_1 A_1} (m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) u + a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} v \\ -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla v] - e^{\alpha_2 A_2} [m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}] v \\ d_1 \frac{\partial u}{\partial \mathbf{n}} - 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} u \\ d_2^0 \frac{\partial v}{\partial \mathbf{n}} \end{bmatrix}.$$

For clarity, we will obtain the local bifurcation in four steps.

Step 1: We claim that the kernel $\text{Ker} \mathcal{L}$ is one-dimensional. In fact, for each $(\varphi, \psi) \in \text{Ker} \mathcal{L}$, we have

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \varphi] - e^{\alpha_1 A_1} (m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \varphi = -a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} \psi & \text{in } \Omega, \\ -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi] - e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi = 0 & \text{in } \Omega, \\ d_1 \frac{\partial \varphi}{\partial \mathbf{n}} = 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} \varphi & \text{on } \partial\Omega, \\ d_2^0 \frac{\partial \psi}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

From the equations of ψ we have $\psi = c\psi_1$ for some constant c , where ψ_1 is the positive eigenfunction associated with the principal eigenvalue d_2^0 . That is, ψ_1 satisfies

$$\begin{cases} -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi] = e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi & \text{in } \Omega, \\ d_2^0 \frac{\partial \psi}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

Then, $\varphi = c\varphi_1$, where $\varphi_1 = \mathcal{L}^{-1}(-a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} \psi_1, 0)^T$ and

$$\mathcal{L} = \begin{bmatrix} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla] - e^{\alpha_1 A_1} (m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \\ d_1 \frac{\partial}{\partial \mathbf{n}} - 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} \end{bmatrix}.$$

Therefore, $\text{Ker} \mathcal{L} = \text{span}\{(\varphi_1, \psi_1)\}$ and hence $\text{Ker} \mathcal{L}$ is one-dimensional. Here $\mathcal{L}$ is invertible since

$$\begin{aligned} & \mu_1(d_1, m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}, 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \\ & < \mu_1(d_1, m_1 - e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}, q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) = 0. \end{aligned}$$

Step 2: We claim that the codimension of $\text{Range} \mathcal{L}$ is 1. In fact, if $(y_{11}, y_{12}, y_{21}, y_{22})^T \in \text{Range} \mathcal{L}$, then there exists $(\varphi, \psi) \in \mathbb{X}^2$ such that

$$\begin{cases} y_{11} = -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \varphi] - e^{\alpha_1 A_1} (m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \varphi + a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} \psi & \text{in } \Omega, \\ y_{12} = -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi] - e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi & \text{in } \Omega, \\ y_{21} = d_1 \frac{\partial \varphi}{\partial \mathbf{n}} - 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} \varphi & \text{on } \partial\Omega, \\ y_{22} = d_2^0 \frac{\partial \psi}{\partial \mathbf{n}} & \text{on } \partial\Omega. \end{cases}$$

For the equations for ψ , we have

$$\int_{\Omega} \psi_1 y_{12} dx = \int_{\partial\Omega} \psi_1 y_{22} ds.$$

In this situation, the equations for φ admits the solution

$$\hat{\varphi}_1 = \mathcal{L}^{-1}(y_{11} - a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} \psi, y_{21})^T.$$

This means the codimension of $\text{Range} \mathcal{L}$ is 1.

Step 3: We shall show that the transversality condition, that is,

$$\mathcal{O}_{(u,v), d_2}(d_2^0, \theta_{d_1, m_1, q_1}, 0) \begin{pmatrix} \varphi_1 \\ \psi_1 \end{pmatrix} \notin \text{Range} \mathcal{L}. \quad (6.2)$$

A straightforward calculations yield

$$\mathcal{O}_{(u,v), d_2}(d_2^0, \theta_{d_1, m_1, q_1}, 0) \begin{pmatrix} \varphi_1 \\ \psi_1 \end{pmatrix} = (d_2^0)^{-1} \begin{pmatrix} 0 \\ -e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi_1 \\ 0 \\ 0 \end{pmatrix}.$$

If (6.2) is false, then there exists $(\varphi, \psi) \in \mathbb{X}^2$ such that

$$\begin{cases} 0 = -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \varphi] - e^{\alpha_1 A_1} (m_1 - 2e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \varphi \\ \quad + a e^{\alpha_1 A_1 + \alpha_2 A_2} \theta_{d_1, m_1, q_1} \psi & \text{in } \Omega, \\ 0 = -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi] - e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi \\ \quad + (d_2^0)^{-1} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi_1 & \text{in } \Omega, \\ 0 = d_1 \frac{\partial \varphi}{\partial \mathbf{n}} - 2q_1 e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1} \varphi & \text{on } \partial\Omega, \\ 0 = d_2 \frac{\partial \psi}{\partial \mathbf{n}} & \text{on } \partial\Omega. \end{cases}$$

Then, as Step 2, we have

$$\int_{\Omega} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi_1^2 dx = 0.$$

This contradicts the condition (2.2). Consequently, all the assumptions for applying the local bifurcation theorem of Crandall and Rabinowitz in [45] have been checked, and thus, the local bifurcation curve $(d_2(s), u(s), v(s))$ is obtained and has the following form

$$(d_2(s), u(s), v(s)) = \left(d_2^0 + \zeta(s), \theta_{d_1, m_1, q_1} + s\varphi_1 + s\varphi(s), s\psi_1 + s\psi(s) \right) \quad (6.3)$$

for $s \in (-\varepsilon, \varepsilon)$, where $(\zeta(s), \varphi(s), \psi(s)) \in \mathbb{R} \times \mathbb{X}^2$ is continuously differentiable for $s \in (-\varepsilon, \varepsilon)$ and satisfies $(\zeta(0), \varphi(0), \psi(0)) = (0, 0, 0)$.

Step 4: Calculation $\zeta'(0)$. It is clear that $(d_2(s), u(s), v(s))$ satisfies

$$\begin{cases} 0 = -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u(s)] - e^{\alpha_1 A_1} u(s) [m_1 - e^{\alpha_1 A_1} u(s) - a e^{\alpha_2 A_2} v(s)] & \text{in } \Omega, \\ 0 = -d_2(s) \nabla \cdot [e^{\alpha_2 A_2} \nabla v(s)] - e^{\alpha_2 A_2} v(s) [m_2 - e^{\alpha_2 A_2} v(s) + b e^{\alpha_1 A_1} u(s)] & \text{in } \Omega, \\ 0 = d_1 \frac{\partial u(s)}{\partial \mathbf{n}} - q_1 e^{\alpha_1 A_1} u^2(s) & \text{on } \partial\Omega, \\ 0 = d_2(s) \frac{\partial v(s)}{\partial \mathbf{n}} - q_2 e^{\alpha_2 A_2} v^2(s) & \text{on } \partial\Omega. \end{cases} \quad (6.4)$$

Multiplying the equation of $v(s)$ by ψ_1 and integrating the result yields

$$\begin{aligned} d_2(s) \int_{\Omega} e^{\alpha_2 A_2} \nabla \psi_1 \nabla v(s) dx &= \int_{\Omega} e^{\alpha_2 A_2} [m_2 - e^{\alpha_2 A_2} v(s) + b e^{\alpha_1 A_1} u(s)] \psi_1 v(s) dx \\ &\quad + \int_{\partial\Omega} q_2 e^{2\alpha_2 A_2} v^2(s) \psi_1 ds. \end{aligned} \quad (6.5)$$

In view of $d_2^0 = v_1(e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}))$, that is, d_2^0 satisfies

$$\begin{cases} -d_2^0 \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi_1] = e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi_1 & \text{in } \Omega, \\ d_2^0 \frac{\partial \psi_1}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

Multiplying the above equation by $v(s)$ and integrating the result gives

$$d_2^0 \int_{\Omega} e^{\alpha_2 A_2} \nabla \psi_1 \nabla v(s) dx = \int_{\Omega} e^{\alpha_2 A_2} [m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}] \psi_1 v(s) dx. \quad (6.6)$$

It follows from (6.5) and (6.6) that

$$\begin{aligned} &\zeta(s) \int_{\Omega} e^{\alpha_2 A_2} [m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}] \psi_1 v(s) dx \\ &= d_2^0 \left[\int_{\Omega} e^{\alpha_2 A_2} \psi_1 v(s) (b e^{\alpha_1 A_1} (u(s) - \theta_{d_1, m_1, q_1}) - e^{\alpha_2 A_2} v(s)) dx + \int_{\partial\Omega} q_2 e^{2\alpha_2 A_2} \psi_1 v^2(s) ds \right]. \end{aligned}$$

Dividing the above equation by s^2 , then letting $s \rightarrow 0^+$ and using (6.3), we have

$$\zeta'(0) = \frac{d_2^0 \left[\int_{\Omega} e^{\alpha_2 A_2} \psi_1^2 (b e^{\alpha_1 A_1} \varphi_1 - e^{\alpha_2 A_2} \psi_1) dx + \int_{\partial\Omega} q_2 e^{2\alpha_2 A_2} \psi_1^3 ds \right]}{\int_{\Omega} e^{\alpha_2 A_2} (m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}) \psi_1^2 dx}. \quad (6.7)$$

We summarize the above discussion as follows.

Theorem 6.1. Under conditions (A1), (A2), (H1), and (2.2), positive solutions of (5.1) emanate from Γ_u at $d_2 = d_2^0$. More precisely, there is a small neighborhood $\mathcal{N}_1$ of $(d_2, u, v) = (d_2^0, \theta_{d_1, m_1, q_1}, 0)$ in $\mathbb{R}_+ \times \mathbb{X}^2$ such that $\mathcal{O}^{-1}(0) \cap \mathcal{N}_1$ consists of the union of $\Gamma_u \cap \mathcal{N}_1$ and the local curve (6.3). Moreover $\zeta'(0)$ is given by (6.7). Therefore, positive solutions contained in $\mathcal{O}^{-1}(0) \cap \mathcal{N}_1$ can be expressed as

$$S_1 = \left\{ \left(d_2^0 + \zeta(s), \theta_{d_1, m_1, q_1} + s\varphi_1 + s\varphi(s), s\psi_1 + s\psi(s) \right) : s \in (0, \varepsilon) \right\}. \quad (6.8)$$

We next analyze the steady-state bifurcation from the branch Γ_v at $d_2 = d_2^*$, where d_2^* is as defined in Lemma 4.4, and this is where the stability of Γ_v changes according to Lemma 4.4 under the condition (2.4). Let φ_2 be the positive eigenfunction associated with $v_1(e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2}))$, and set $\psi_2 = \mathcal{T}^{-1}(be^{\alpha_1 A_1 + \alpha_2 A_2}\theta_{d_2^*, m_2, q_2}\varphi_2, 0)^T$ and

$$\mathcal{T} = \begin{bmatrix} -d_2^* \nabla \cdot [e^{\alpha_2 A_2} \nabla] - e^{\alpha_2 A_2}(m_2 - 2e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2}) \\ d_2^* \frac{\partial}{\partial \mathbf{n}} - 2q_2 e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2} \end{bmatrix}.$$

The invertibility of operator $\mathcal{T}$ is analogous to that of $\mathcal{L}$. Thus, similarly to Theorem 6.1, we have

Theorem 6.2. In addition to conditions (A1), (A2), (H2), and (2.4), assume that θ_{d_2, m_2, q_2} is locally monotonic near $d_2 = d_2^*$ and

$$\int_{\Omega} e^{\alpha_1 A_1} m_1 dx < 0.$$

Then positive solutions of (5.1) emanate from Γ_v at $d_2 = d_2^*$. More precisely, there is a small neighborhood $\mathcal{N}_2$ of $(d_2, u, v) = (d_2^*, 0, \theta_{d_2^*, m_2, q_2})$ in $\mathbb{R}_+ \times \mathbb{X}^2$ such that $\mathcal{O}^{-1}(0) \cap \mathcal{N}_2$ consists of the union of $\Gamma_v \cap \mathcal{N}_2$ and the local curve

$$(d_2(s), u(s), v(s)) = (d_2^* + \kappa(s), s\varphi_2 + s\varphi(s), \theta_{d_2^*, m_2, q_2} + s\psi_2 + s\psi(s)) \quad (6.9)$$

for $s \in (-\varepsilon, \varepsilon)$, where $(\kappa(s), \varphi(s), \psi(s)) \in \mathbb{R} \times \mathbb{X}^2$ is continuously differentiable for $s \in (-\varepsilon, \varepsilon)$ and satisfies $(\kappa(0), \varphi(0), \psi(0)) = (0, 0, 0)$. Therefore, positive solutions contained in $\mathcal{O}^{-1}(0) \cap \mathcal{N}_2$ can be expressed as

$$S_2 = \left\{ (d_2^* + \kappa(s), s\varphi_2 + s\varphi(s), \theta_{d_2^*, m_2, q_2} + s\psi_2 + s\psi(s)) : s \in (0, \varepsilon) \right\}. \quad (6.10)$$

The proof of Theorem 6.2 is very similar to that of Theorem 6.1, and hence we will only sketch it here.

Proof of Theorem 6.2. By straightforward calculations, the linearized operator $\mathcal{P}$ of $\mathcal{O}$ around $(d_2^*, 0, \theta_{d_2^*, m_2, q_2})$ is given by

$$\mathcal{P}(u, v) = \begin{bmatrix} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u] - e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2})u \\ -d_2^* \nabla \cdot [e^{\alpha_2 A_2} \nabla v] - e^{\alpha_2 A_2}[(m_2 - 2e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2})v + be^{\alpha_1 A_1}\theta_{d_2^*, m_2, q_2}u] \\ d_1 \frac{\partial u}{\partial \mathbf{n}} \\ d_2^* \frac{\partial v}{\partial \mathbf{n}} - 2q_2 e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2}v \end{bmatrix}.$$

Now we determine $\text{Ker } \mathcal{P}$. If $(\varphi, \psi) \in \text{Ker } \mathcal{P}$, then

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \varphi] - e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2})\varphi = 0 & \text{in } \Omega, \\ -d_2^* \nabla \cdot [e^{\alpha_2 A_2} \nabla \psi] - e^{\alpha_2 A_2}(m_2 - 2e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2})\psi = be^{\alpha_1 A_1 + \alpha_2 A_2}\theta_{d_2^*, m_2, q_2}\varphi & \text{in } \Omega, \\ d_1 \frac{\partial \varphi}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega, \\ d_2^* \frac{\partial \psi}{\partial \mathbf{n}} - 2q_2 e^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2}\psi = 0 & \text{on } \partial\Omega. \end{cases}$$

By the definition of d_2^* , one has $d_1 = d_1^* = v_1(e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2}))$. Thus $\varphi = c\varphi_2$ for some constant c , and then $\psi = c\psi_2$. We therefore show that $\text{Ker } \mathcal{P} = \text{span}\{(\varphi_2, \psi_2)\}$, and thereby, $\text{Ker } \mathcal{P}$ is one-dimensional.

Similar to the argument in Step 2 of Theorem 6.1, $(y_{11}, y_{12}, y_{21}, y_{22})^T \in \text{Range } \mathcal{P}$ if and only if

$$\int_{\Omega} \varphi_2 y_{11} dx = \int_{\partial\Omega} \varphi_2 y_{21} ds.$$

Thus, the codimension of $\text{Range } \mathcal{P}$ is 1.

We now verify the transversality condition, that is,

$$\mathcal{O}_{(u,v), d_2}(d_2^*, 0, \theta_{d_2^*, m_2, q_2}) \begin{pmatrix} \varphi_2 \\ \psi_2 \end{pmatrix} \notin \text{Range } \mathcal{P}. \quad (6.11)$$

If (6.11) is false, let $w^* = \frac{\partial \theta_{d_2, m_2, q_2}}{\partial d_2} \Big|_{d_2=d_2^*}$, then there exists $(\varphi, \psi) \in \mathbb{X}^2$ such that

$$\begin{cases} -d_1^* \nabla \cdot [e^{\alpha_2 A_2} \nabla \varphi] - e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2}\theta_{d_2^*, m_2, q_2})\varphi = ae^{\alpha_1 A_1} w^* \varphi_2 & \text{in } \Omega, \\ d_1^* \frac{\partial \varphi}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

Then, similar to Step 3 of Theorem 6.1, we have

$$\int_{\Omega} ae^{\alpha_1 A_1} w^* \varphi_2^2 dx = 0. \quad (6.12)$$

We next demonstrate that (6.12) is not true by showing that $w^* > 0$ (or $w^* < 0$). Recall that θ_{d_2, m_2, q_2} satisfies

$$\begin{cases} -d_2 \nabla \cdot [e^{\alpha_2 A_2} \nabla \theta_{d_2, m_2, q_2}] = e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2} [m_2 - e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}] & \text{in } \Omega, \\ d_2 \frac{\partial \theta_{d_2, m_2, q_2}}{\partial \mathbf{n}} = q_2 e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}^2 & \text{on } \partial\Omega. \end{cases}$$

It follows from

$$\begin{aligned} & \mu_1(d_2, m_2 - 2e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, 2q_2 e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) \\ & < \mu_1(d_2, m_2 - e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}, q_2 e^{\alpha_2 A_2} \theta_{d_2, m_2, q_2}) = 0 \end{aligned}$$

that w^* solves

$$\begin{cases} -d_2^* \nabla \cdot [e^{\alpha_2 A_2} \nabla w^*] - e^{\alpha_2 A_2} [m_2 - 2e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}] w^* \\ \quad = -(d_2^*)^{-1} e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2} [m_2 - e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}] & \text{in } \Omega, \\ d_2^* \frac{\partial w^*}{\partial n} - 2q_2 e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2} w^* = -q_2 e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}^2 & \text{on } \partial\Omega. \end{cases}$$

It is easy to see that $w^* \neq 0$, and thus $w^* > 0$ (or $w^* < 0$) since θ_{d_2, m_2, q_2} is locally monotonic near $d_2 = d_2^*$. We thus end the proof of [Theorem 6.2](#) by the local bifurcation theorem of Crandall and Rabinowitz in [45]. $\square$

7. Global bifurcation

Let S denote the closure of the set of all positive steady-state solutions (d_2, u, v) of system (5.1) in the space $\mathbb{R}_+ \times \mathbb{X}^2$, that is, S is the closure of $\{(d_2, u, v) \in \mathbb{R}_+ \times \mathbb{X}^2 : \mathcal{O}(d_2, u, v) = 0, u > 0, v > 0\}$. Obviously, S contains two subcontinua (i.e., a maximal, closed, and connected component): One is denoted by $\tilde{S}_1$ to which $(d_2^0, \theta_{d_1, m_1, q_1}, 0)$ belongs, the other is denoted by $\tilde{S}_2$ to which $(d_2^*, 0, \theta_{d_2^*, m_2, q_2})$ belongs. In particular, $S_1 \subset \tilde{S}_1$ and $S_2 \subset \tilde{S}_2$, where S_1 and S_2 are given in (6.8) and (6.10), respectively. For simplicity, we use S_1 and S_2 to represent $\tilde{S}_1$ and $\tilde{S}_2$, respectively.

We first present the global structure of the solution branch S_1 under the conditions of [Theorem 6.1](#). Note that $\mathbb{X}^2 \hookrightarrow \mathbb{Y}^2$ is compact. Thus, it follows from Theorem 1.2 of [64] that S_1 satisfies one of the following three alternatives:

- (a) S_1 is unbounded in $\mathbb{R}_+ \times \mathbb{X}^2$.
- (b) There exists another eigenvalue of $\mathcal{L}$, say $\bar{d}_2 \neq d_2^0$, such that $(\bar{d}_2, \theta_{d_1, m_1, q_1}, 0) \in S_1$.
- (c) There exists $(d_2, y) \in S_1$ with $y \in \mathbb{Y}_0 \setminus \{0\}$, where $\mathbb{Y}_0$ is the complement of $\text{Ker } \mathcal{L}$ in $\mathbb{X}^2$.

Denote by $\mathbb{P}$ the positive cone in $\mathbb{X}$ whose interior $\text{Int } \mathbb{P}$ is nonempty, and let S_{01} be the subcontinuum of S_1 satisfying

$$S_{01} \subset \mathbb{R}_+ \times \text{Int } \mathbb{P} \times \text{Int } \mathbb{P}.$$

If S_{01} is unbounded, then the projection of S_{01} on d_2 -axis, denoted by $\text{Proj}_{d_2} S_{01}$, is unbounded since any positive solutions (u, v) of system (5.1) are bounded in $\mathbb{X}^2$. In fact, by [Theorem 2.4](#), one has $0 < u < \bar{u}$ and $0 < v < \bar{v}$. Integrating the equations for u and v of system (5.1) on Ω , respectively, one can conclude that ∇u and ∇v are uniformly bounded on $\bar{\Omega}$. Applying the equations for u and v of system (5.1) again, we have Δu and Δv are uniformly bounded on $\bar{\Omega}$. Therefore, any positive solutions of system (5.1) must be bounded in $\mathbb{X}^2$. In view of $d_2 > 0$, $\text{Proj}_{d_2} S_{01} \subset (d_2^0, +\infty)$.

Now we assume that S_{01} is bounded in $\mathbb{R}_+ \times \mathbb{X}^2$. Suppose $S_{01} = S_1$. Then S_1 is bounded, and hence alternative (a) is not satisfied. Suppose that alternative (c) happens. Then there exists

$$(d_2, y) \in S_1 = S_{01} \subset \mathbb{R}_+ \times \text{Int } \mathbb{P} \times \text{Int } \mathbb{P}$$

for some $y = (y_{11}, y_{12}, y_{21}, y_{22}) \in \mathbb{Y}_0 \setminus \{0\}$. In particular,

$$\int_{\Omega} \psi_1 y_{12} dx + \int_{\Omega} \psi_1 y_{22} ds = 0,$$

where ψ_1 is the positive eigenfunction associated with the principal eigenvalue d_2^0 . This contradicts the fact $\int_{\Omega} \psi_1 y_{12} dx + \int_{\Omega} \psi_1 y_{22} ds > 0$. Consequently, S_{01} satisfies alternative (b). That is, there exists another eigenvalue of $\mathcal{L}$, say $\bar{d}_2 \neq d_2^0$, such that $(\bar{d}_2, \theta_{d_1, m_1, q_1}, 0) \in S_{01}$. Hence, there exists a sequence $(d_2^n, u^n, v^n) \subset S_{01}$ such that

$$(d_2^n, u^n, v^n) \rightarrow (\bar{d}_2, \theta_{d_1, m_1, q_1}, 0) \text{ in } \mathbb{R}_+ \times \mathbb{X}^2.$$

Noticing that (d_2^n, u^n, v^n) satisfies

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u^n] = e^{\alpha_1 A_1} u^n [m_1 - e^{\alpha_1 A_1} u^n - a e^{\alpha_2 A_2} v^n] & \text{in } \Omega, \\ -d_2^n \nabla \cdot [e^{\alpha_2 A_2} \nabla v^n] = e^{\alpha_2 A_2} v^n [m_2 - e^{\alpha_2 A_2} v^n + b e^{\alpha_1 A_1} u^n] & \text{in } \Omega, \\ d_1 \frac{\partial u^n}{\partial n} = q_1 e^{\alpha_1 A_1} (u^n)^2 & \text{on } \partial\Omega, \\ d_2^n \frac{\partial v^n}{\partial n} = q_2 e^{\alpha_2 A_2} (v^n)^2 & \text{on } \partial\Omega. \end{cases} \quad (7.1)$$

Define

$$\bar{v}^n := \frac{v^n}{\|v^n\|_{\mathbb{X}}}, \quad n \geq 1.$$

Then $\hat{v}^n$ satisfies

$$\begin{cases} -d_2^n \nabla \cdot [e^{\alpha_2 A_2} \nabla \hat{v}^n] = e^{\alpha_2 A_2} \hat{v}^n [m_2 - e^{\alpha_2 A_2} v^n + b e^{\alpha_1 A_1} u^n] & \text{in } \Omega, \\ d_2^n \frac{\partial \hat{v}^n}{\partial n} = q_2 e^{\alpha_2 A_2} \hat{v}^n v^n & \text{on } \partial\Omega, \end{cases}$$

and is bounded in $\mathbb{X}$. Hence, $\hat{v}^n$ converges to $\hat{v}^\infty$ in $W^{1,p}(\overline{\Omega})$ since $\mathbb{X} = W^{2,p}(\overline{\Omega}) \hookrightarrow W^{1,p}(\overline{\Omega})$ is compact. Letting $n \rightarrow \infty$ we see that $\hat{v}^\infty$ satisfies

$$\begin{cases} -\bar{d}_2 \nabla \cdot [e^{\alpha_2 A_2} \nabla \hat{v}^\infty] = e^{\alpha_2 A_2} \hat{v}^\infty [m_2 + b e^{\alpha_1 A_1} \theta_{d_1, m_1, q_1}] & \text{in } \Omega, \\ \bar{d}_2 \frac{\partial \hat{v}^\infty}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases}$$

In particular, by the elliptic regularity theory, $\hat{v}^\infty \in \mathbb{X}$. Since $\hat{v}^\infty > 0$ and $\|\hat{v}^\infty\|_{\mathbb{X}} = 1$, we have $\bar{d}_2 = d_2^0$ by the uniqueness of the principal eigenvalue, which is a contradiction. This shows that $S_{01} = S_1$ cannot occur. Consequently:

$$S_{01} \subset S_1 \text{ and } S_{01} \neq S_1.$$

Since S_{01} is a proper subset of S_1 , there exist $(d_2^\infty, u^\infty, v^\infty) \in \partial(\mathbb{R}_+ \times \mathbb{P} \times \mathbb{P})$ and a sequence

$$(d_2^n, u^n, v^n) \in \mathbb{R}_+ \times \text{Int}\mathbb{P} \times \text{Int}\mathbb{P}$$

such that

$$(d_2^n, u^n, v^n) \rightarrow (d_2^\infty, u^\infty, v^\infty) \text{ in } \mathbb{R}_+ \times \mathbb{X}^2.$$

By the continuity of $\mathcal{O}$, $0 = \mathcal{O}(d_2^n, u^n, v^n) \rightarrow \mathcal{O}(d_2^\infty, u^\infty, v^\infty)$, that is, $(d_2^\infty, u^\infty, v^\infty)$ is a non-negative solution of (5.1). Moreover, since $(d_2^\infty, u^\infty, v^\infty) \in \partial(\mathbb{R}_+ \times \mathbb{P} \times \mathbb{P})$, we have $(u^\infty, v^\infty) \in \partial(\mathbb{P} \times \mathbb{P})$ when $d_2^\infty \neq 0$. Thus, by the Strong Maximum Principle, either $u^\infty = 0$ or $v^\infty = 0$ if $d_2^\infty \neq 0$. In what follows, we consider the three possible cases for $d_2^\infty \neq 0$:

Case 1: $u^\infty = 0$ and $v^\infty > 0$. Define

$$\hat{u}^n := \frac{u^n}{\|u^n\|_{\mathbb{X}}}, \quad n \geq 1.$$

Consequently, $\hat{u}^n$ is bounded in $\mathbb{X}$, and as before, $\hat{u}^n$ converges to $\hat{u}^\infty$ in $W^{1,p}(\overline{\Omega})$. Recall that $\mathcal{O}(d_2^n, u^n, v^n) = 0$, then $(d_2^n, \hat{u}^n, v^n)$ satisfies

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \hat{u}^n] = e^{\alpha_1 A_1} \hat{u}^n [m_1 - e^{\alpha_1 A_1} u^n - a e^{\alpha_2 A_2} v^n] & \text{in } \Omega, \\ -d_2^n \nabla \cdot [e^{\alpha_2 A_2} \nabla v^n] = e^{\alpha_2 A_2} v^n [m_2 - e^{\alpha_2 A_2} v^n + b e^{\alpha_1 A_1} u^n] & \text{in } \Omega, \\ d_1 \frac{\partial \hat{u}^n}{\partial n} = q_1 e^{\alpha_1 A_1} \hat{u}^n u^n & \text{on } \partial\Omega, \\ d_2^n \frac{\partial v^n}{\partial n} = q_2 e^{\alpha_2 A_2} (v^n)^2 & \text{on } \partial\Omega. \end{cases} \quad (7.2)$$

Letting $n \rightarrow \infty$ yields

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla \hat{u}^\infty] = e^{\alpha_1 A_1} \hat{u}^\infty [m_1 - a e^{\alpha_2 A_2} v^\infty] & \text{in } \Omega, \\ d_1 \frac{\partial \hat{u}^\infty}{\partial n} = 0 & \text{on } \partial\Omega, \end{cases} \quad (7.3)$$

and

$$\begin{cases} -d_2^\infty \nabla \cdot [e^{\alpha_2 A_2} \nabla v^\infty] = e^{\alpha_2 A_2} v^\infty [m_2 - e^{\alpha_2 A_2} v^\infty] & \text{in } \Omega, \\ d_2^\infty \frac{\partial v^\infty}{\partial n} = q_2 e^{\alpha_2 A_2} (v^\infty)^2 & \text{on } \partial\Omega. \end{cases} \quad (7.4)$$

It follows from (7.4) that $(d_2^\infty, 0, v^\infty)$ is a semi-trivial solution of system (5.1) and hence $v^\infty = \theta_{d_2^\infty, m_2, q_2}$. And for Eq. (7.3), we have

$$d_1 = v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} v^\infty)) = v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$$

since $\hat{u}^\infty > 0$. That is, there exists a semi-trivial solution $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2})$ of (5.1) such that $d_1 = v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$ and $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2}) \in \overline{S_{01}}$. On the other hand, if $\int_\Omega e^{\alpha_1 A_1} m_1 dx \geq 0$, then $v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2})) < 0$, a contradiction with $d_1 > 0$. We thus suppose that $\int_\Omega e^{\alpha_1 A_1} m_1 dx < 0$, then $\mathbb{D}_1 = (0, v_1(e^{\alpha_1 A_1} m_1))$. Therefore,

$$v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \bar{v})) < v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} v^\infty)) < v_1(e^{\alpha_1 A_1} m_1)$$

since $\bar{v} > v^\infty > 0$, and by Lemma 3.1, there exist $\delta_1, \delta_2 > 0$ such that

$$v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \bar{v})) + \delta_1 \leq v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} v^\infty)) \leq v_1(e^{\alpha_1 A_1} m_1) - \delta_2.$$

This means that Case 1 does not occur if $d_1 \in (0, v_1(e^{\alpha_1 A_1} (m_1 - a e^{\alpha_2 A_2} \bar{v})) + \delta_1) \cup (v_1(e^{\alpha_1 A_1} m_1) - \delta_2, v_1(e^{\alpha_1 A_1} m_1))$ when $\int_\Omega e^{\alpha_1 A_1} m_1 dx < 0$ or if $\int_\Omega e^{\alpha_1 A_1} m_1 dx \geq 0$.

Case 2: $u^\infty > 0$ and $v^\infty = 0$. Define

$$\hat{v}^n := \frac{v^n}{\|v^n\|_{\mathbb{X}}}, \quad n \geq 1.$$

Then $\hat{v}^n$ is bounded in $\mathbb{X}$, and hence $\hat{v}^n$ converges to $\hat{v}^\infty$ in $W^{1,p}(\bar{\Omega})$ by the same way of Case 1. Also, similar to Case 1, we have

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u^\infty] = e^{\alpha_1 A_1} u^\infty [m_1 - e^{\alpha_1 A_1} u^\infty] & \text{in } \Omega, \\ d_1 \frac{\partial u^\infty}{\partial n} = q_1 e^{\alpha_1 A_1} (u^\infty)^2 & \text{on } \partial\Omega, \end{cases} \quad (7.5)$$

and

$$\begin{cases} -d_2^\infty \nabla \cdot [e^{\alpha_2 A_2} \nabla \hat{v}^\infty] = e^{\alpha_2 A_2} \hat{v}^\infty [m_2 + b e^{\alpha_1 A_1} u^\infty] & \text{in } \Omega, \\ d_2^\infty \frac{\partial \hat{v}^\infty}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases} \quad (7.6)$$

It follows from (7.5) that $(d_2^\infty, u^\infty, 0)$ is a semi-trivial solution of system (5.1). And for the Eq. (7.6), $d_2^\infty = v_1(e^{\alpha_2 A_2}(m_2 + b e^{\alpha_1 A_1} u^\infty))$ since $\hat{v}^\infty > 0$. On the other hand, by construction, there exists $\delta > 0$ such that $(d_2^\infty, u^\infty, 0) \in S_1 \setminus B_\delta(d_2^\infty, \theta_{d_1, m_1, q_1}, 0)$. Indeed, for $\delta > 0$ small enough we have $S_{01} = S_1$ in $B_\delta(d_2^\infty, \theta_{d_1, m_1, q_1}, 0)$. In particular, $(d_2^\infty, u^\infty, 0) \neq (d_2^0, \theta_{d_1, m_1, q_1}, 0)$. By the uniqueness of the positive steady-state solution of Eq. (3.2), however, we have $u^\infty = \theta_{d_1, m_1, q_1}$ and so $d_2^\infty = d_2^0$. Hence, $(d_2^\infty, u^\infty, 0) = (d_2^0, \theta_{d_1, m_1, q_1}, 0)$, a contradiction.

Case 3: $u^\infty = 0$ and $v^\infty = 0$. Define

$$\hat{u}^n := \frac{u^n}{\|u^n\|_{\mathbb{X}}}, \quad \hat{v}^n := \frac{v^n}{\|v^n\|_{\mathbb{X}}}, \quad n \geq 1.$$

Similar to Case 1, we have

$$\hat{u}^n \rightarrow \hat{u}^\infty \text{ in } W^{1,p}(\bar{\Omega}),$$

$$\hat{v}^n \rightarrow \hat{v}^\infty \text{ in } W^{1,p}(\bar{\Omega}),$$

and $\hat{u}^\infty$ satisfies (7.3) with $v^\infty = 0$ and $\hat{v}^\infty$ satisfies (7.6) with $u^\infty = 0$. Thus, $d_1 = v_1(e^{\alpha_1 A_1} m_1)$ and $d_2^\infty = v_1(e^{\alpha_2 A_2} m_2)$, a contradiction with $d_1 > 0$ when $\int_\Omega e^{\alpha_1 A_1} m_1 dx \geq 0$ or with $d_1 < v_1(e^{\alpha_1 A_1} m_1)$ when $\int_\Omega e^{\alpha_1 A_1} m_1 dx < 0$.

We now consider the case where $d_2^\infty = 0$. Note that $d_2^\infty \neq 0$ in Case 2 and Case 3, then we only need to consider:

Case 4: $d_2^\infty = 0$, $v^\infty > 0$. For (7.1), let $n \rightarrow \infty$, we have

$$\begin{cases} -d_2^\infty \nabla \cdot [e^{\alpha_2 A_2} \nabla v^\infty] = e^{\alpha_2 A_2} v^\infty [m_2 - e^{\alpha_2 A_2} v^\infty + b e^{\alpha_1 A_1} u^\infty] & \text{in } \Omega, \\ d_2^\infty \frac{\partial v^\infty}{\partial n} = q_2 e^{\alpha_2 A_2} (v^\infty)^2 & \text{on } \partial\Omega. \end{cases} \quad (7.7)$$

It is not difficult to verify that Eq. (7.7) has no positive solution when $d_2^\infty = 0$, which means that Case 4 cannot occur.

We thus summarize the above discussion as follows.

Theorem 7.1. Under the condition of Theorem 6.1, from the point $(d_2, u, v) = (d_2^0, \theta_{d_1, m_1, q_1}, 0)$ emanates a continuum

$$S_{01} \subset \mathbb{R}_+ \times \text{Int}\mathbb{P} \times \text{Int}\mathbb{P}$$

of positive solution of (5.1) satisfying that either:

- (i) $\text{Proj}_{d_2} S_{01} \supset (d_2^0, +\infty)$; or
- (ii) there exists a semi-trivial solution $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2})$ of (5.1) such that $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2}) \in \overline{S_{01}}$, where d_2^∞ is uniquely determined by $d_1 = v_1(e^{\alpha_1 A_1}(m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$.

Furthermore, (ii) does not occur if

$$d_1 \in (0, v_1(e^{\alpha_1 A_1}(m_1 - a e^{\alpha_2 A_2} \bar{v})) + \delta_1) \cup (v_1(e^{\alpha_1 A_1} m_1) - \delta_2, v_1(e^{\alpha_1 A_1} m_1))$$

when $\int_\Omega e^{\alpha_1 A_1} m_1 dx < 0$ or if $\int_\Omega e^{\alpha_1 A_1} m_1 dx \geq 0$. Thus, (5.1) has at least one positive solution for $d_2 > d_2^0$ if $d_1 \in (0, v_1(e^{\alpha_1 A_1}(m_1 - a e^{\alpha_2 A_2} \bar{v})) + \delta_1) \cup (v_1(e^{\alpha_1 A_1} m_1) - \delta_2, v_1(e^{\alpha_1 A_1} m_1))$ when $\int_\Omega e^{\alpha_1 A_1} m_1 dx < 0$ or if $\int_\Omega e^{\alpha_1 A_1} m_1 dx \geq 0$, where $\bar{v}$ is as defined in (2.6).

The global structure of the solution branch S_2 under the conditions of Theorem 6.2, is analogous as follows.

Theorem 7.2. Under the condition of Theorem 6.2, from the point $(d_2, u, v) = (d_2^*, 0, \theta_{d_2^*, m_2, q_2})$ emanates a continuum

$$S_{02} \subset \mathbb{D}_2 \times \text{Int}\mathbb{P} \times \text{Int}\mathbb{P}$$

of positive solution of (5.1). In addition, if $d_1 \in (v_1(e^{\alpha_1 A_1}(m_1 - a m_2^+)), v_1(e^{\alpha_1 A_1} m_1))$, then there exists a semi-trivial solution $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2})$ of (5.1) such that $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2}) \in \overline{S_{02}}$, where d_2^∞ is uniquely determined by $d_1 = v_1(e^{\alpha_1 A_1}(m_1 - a e^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$. Furthermore $d_2^\infty = \hat{d}_2$ when $\theta_{d_2^\infty, m_2, q_2} = \theta_{\hat{d}_2^*, m_2, q_2}$, where $\hat{d}_2^* \neq d_2^*$ is another eigenvalue of $\mathcal{P}$.

Thus, (5.1) has at least one positive solution when $d_1 \in (v_1(e^{\alpha_1 A_1}(m_1 - a m_2^+)), v_1(e^{\alpha_1 A_1} m_1))$ and d_2 is between d_2^∞ and d_2^* , where the relative magnitudes of d_2^∞ and d_2^* remain undetermined.

Proof. In view of Lemma 4.4, d_2^* is uniquely determined by (2.5), that is, $d_1 = v_1(e^{\alpha_1 A_1}(m_1 - a e^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}))$. Therefore, d_2^* exists only if d_1 lies within the region $(v_1(e^{\alpha_1 A_1}(m_1 - a m_2^+)), v_1(e^{\alpha_1 A_1} m_1))$, and so we constrain d_1 to this parameter regime throughout the subsequent analysis.

Similar to Theorem 7.1, it follows from Theorem 1.2 of [64] that S_2 satisfies one of the following three alternatives:

- (d) S_2 is unbounded in $\mathbb{D}_2 \times \mathbb{X}^2$.
- (e) There exists another eigenvalue of $\mathcal{P}$, say $\hat{d}_2 \neq d_2^*$, such that $(\hat{d}_2, 0, \theta_{d_2^*, m_2, q_2}) \in S_2$.
- (f) There exists $(d_2, z) \in S_2$ with $z \in \mathbb{Y}_1 \setminus \{0\}$, where $\mathbb{Y}_1$ is the complement of $\text{Ker } \mathcal{P}$ in $\mathbb{X}^2$.

Let S_{02} be the subcontinuum of S_2 satisfying

$$S_{02} \subset \mathbb{D}_2 \times \text{Int } \mathbb{P} \times \text{Int } \mathbb{P}.$$

If S_{02} is unbounded, then positive solutions of system (5.1) are unbounded in $\mathbb{X}^2$ by the boundedness of $\mathbb{D}_2$. This is impossible because any positive solutions of system (5.1) must be unbounded in $\mathbb{X}^2$ as shown in the argument of Theorem 7.1. Thus S_{02} is bounded in $\mathbb{D}_2 \times \mathbb{X}^2$.

Suppose $S_{02} = S_2$. Then S_2 is bounded and, hence, alternative (d) is not satisfied. Suppose that alternative (f) happens. Then there exists

$$(d_2, z) \in S_2 = S_{02} \subset \mathbb{D}_2 \times \text{Int } \mathbb{P} \times \text{Int } \mathbb{P}$$

for some $z = (z_{11}, z_{12}, z_{21}, z_{22}) \in \mathbb{Y}_1 \setminus \{0\}$. In particular,

$$\int_{\Omega} \varphi_2 z_{11} dx + \int_{\Omega} \varphi_2 z_{21} ds = 0,$$

where φ_2 is the positive eigenfunction associated with the principal eigenvalue $\nu_1(e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2} \theta_{d_2^*, m_2, q_2}))$. This contradicts the fact $\int_{\Omega} \varphi_2 z_{11} dx + \int_{\Omega} \varphi_2 z_{21} ds > 0$. Consequently, S_{02} satisfies alternative (e). That is, there exists another eigenvalue of $\mathcal{P}$, say $\hat{d}_2 \neq d_2^*$, such that $(\hat{d}_2, 0, \theta_{d_2^*, m_2, q_2}) \in S_{02}$.

Suppose $S_{02} \subset S_2$ and $S_{02} \neq S_2$. Since S_{02} is a proper subset of S_2 , there exist $(d_2^\infty, u^\infty, v^\infty) \in \partial(\mathbb{D}_2 \times \mathbb{P} \times \mathbb{P})$ and a sequence

$$(d_2^n, u^n, v^n) \in \mathbb{D}_2 \times \text{Int } \mathbb{P} \times \text{Int } \mathbb{P}$$

such that

$$(d_2^n, u^n, v^n) \rightarrow (d_2^\infty, u^\infty, v^\infty) \text{ in } \mathbb{D}_2 \times \mathbb{X}^2.$$

By the continuity of $\mathcal{O}$, $0 = \mathcal{O}(d_2^n, u^n, v^n) \rightarrow \mathcal{O}(d_2^\infty, u^\infty, v^\infty)$, that is, $(d_2^\infty, u^\infty, v^\infty)$ is a non-negative solution of (5.1). Moreover, since $(d_2^\infty, u^\infty, v^\infty) \in \partial(\mathbb{D}_2 \times \mathbb{P} \times \mathbb{P})$, we have $(u^\infty, v^\infty) \in \partial(\mathbb{P} \times \mathbb{P})$ when $d_2^\infty \notin \{0, \nu_1(e^{\alpha_2 A_2} m_2)\}$. Thus, by the Strong Maximum Principle, either $u^\infty = 0$ or $v^\infty = 0$ if $d_2^\infty \notin \{0, \nu_1(e^{\alpha_2 A_2} m_2)\}$. We first consider the three possible cases for $d_2^\infty \notin \{0, \nu_1(e^{\alpha_2 A_2} m_2)\}$:

Case 5: $u^\infty = 0$ and $v^\infty > 0$. This case is exactly the same as Case 1 in Theorem 7.1, that is, there exists a semi-trivial solution $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2})$ of (5.1) such that $d_1 = \nu_1(e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2} \theta_{d_2^\infty, m_2, q_2}))$ and $(d_2^\infty, 0, \theta_{d_2^\infty, m_2, q_2}) \in \overline{S_{02}}$. This means that Case 5 coincides with (e).

Case 6: $u^\infty > 0$ and $v^\infty = 0$. The same discussion as in Case 2 yields $d_2^\infty = \nu_1(e^{\alpha_2 A_2}(m_2 + be^{\alpha_1 A_1} u^\infty))$. It follows from $u^\infty > 0$ and Lemma 3.3 that $d_2^\infty > \nu_1(e^{\alpha_2 A_2} m_2)$, which contradicts $d_2^\infty \in \mathbb{D}_2$.

Case 7: $u^\infty = 0$ and $v^\infty = 0$. The same discussion as in Case 3 gives $d_1 = \nu_1(e^{\alpha_1 A_1} m_1)$ and $d_2^\infty = \nu_1(e^{\alpha_2 A_2} m_2)$, a contradiction with $d_2^\infty \notin \{0, \nu_1(e^{\alpha_2 A_2} m_2)\}$.

Now we consider the case where $d_2^\infty \in \{0, \nu_1(e^{\alpha_2 A_2} m_2)\}$. Note that the case where $d_2^\infty = 0$ is exactly the same as Case 4 and cannot occur. In addition, $d_2^\infty \neq \nu_1(e^{\alpha_2 A_2} m_2)$ in Case 6, and the conclusion in Case 5 does not depend on the value of d_2^∞ . We thus consider the two possible cases for $d_2^\infty = \nu_1(e^{\alpha_2 A_2} m_2)$:

Case 8: $d_2^\infty = \nu_1(e^{\alpha_2 A_2} m_2)$, $u^\infty = 0$ and $v^\infty = 0$. From the discussion of Case 7, we know that $d_1 = \nu_1(e^{\alpha_1 A_1} m_1)$, a contradiction with $d_1 \in (\nu_1(e^{\alpha_1 A_1}(m_1 - am_2^+)), \nu_1(e^{\alpha_1 A_1} m_1))$.

Case 9: $d_2^\infty = \nu_1(e^{\alpha_2 A_2} m_2)$, $u^\infty > 0$ and $v^\infty > 0$. For (7.1), let $n \rightarrow \infty$, then u^∞ satisfies

$$\begin{cases} -d_1 \nabla \cdot [e^{\alpha_1 A_1} \nabla u^\infty] = e^{\alpha_1 A_1} u^\infty [m_1 - e^{\alpha_1 A_1} u^\infty - ae^{\alpha_2 A_2} v^\infty] & \text{in } \Omega, \\ d_1 \frac{\partial u^\infty}{\partial n} = q_1 e^{\alpha_1 A_1} (u^\infty)^2 & \text{on } \partial \Omega, \end{cases} \quad (7.8)$$

and v^∞ satisfies (7.7). It follows from $\int_{\Omega} e^{\alpha_1 A_1} m_1 dx < 0$ that $\int_{\Omega} e^{\alpha_1 A_1} (m_1 - ae^{\alpha_2 A_2} v^\infty) dx < 0$. Therefore, by Lemma 3.4, problem (7.8) has a positive solution u^∞ if and only if $\nu_1(e^{\alpha_1 A_1}(m_1 - am_2^+)) < d_1 < \nu_1(e^{\alpha_1 A_1}(m_1 - ae^{\alpha_2 A_2} v^\infty))$, which means that $e^{\alpha_2 A_2} v^\infty < m_2^+$. This is impossible since $v^\infty > 0$. The proof is completed. $\square$

8. Conclusion and discussion

This paper presents a complete analytical framework for understanding global bifurcation structures in predator-prey systems with nonlinear boundary conditions, where density-dependent dispersal at habitat edges creates fundamentally new dynamical phenomena. By developing a unified approach that treats the predator's diffusion coefficient d_2 as the primary bifurcation parameter, we have successfully combined Crandall-Rabinowitz local bifurcation theory with global continuation techniques to obtain a comprehensive characterization of coexistence states. Our analysis reveals that smooth branches of positive solutions emerge precisely at critical values d_2^0 and d_2^* where semi-trivial solutions undergo stability changes, demonstrating how spatial heterogeneity mediates species interactions through the boundary conditions.

The global extension of these local bifurcations via unilateral bifurcation theory shows that solution branches exhibit two distinct behaviors: either persisting for all diffusion rates beyond critical thresholds or forming closed loops that reconnect to other semi-trivial states. This complete topological description of the solution set highlights how nonlinear boundary terms $q_1 u^2$ and $q_2 v^2$ introduce qualitatively new features compared to classical models with linear boundary conditions. Ecologically, these mathematical results provide mechanistic explanations for edge effects in fragmented habitats, showing how density-dependent dispersal can create stability windows at intermediate predator mobility ranges while preventing coexistence at both extremely low and high dispersal rates.

While our study provides a rigorous foundation for analyzing nonlinear boundary value problems in ecology, it naturally leads to several important open questions. The extension of these results to multi-species communities presents both mathematical challenges and opportunities for deeper ecological insights, particularly in understanding how boundary effects influence food web stability. Numerical verification of the Γ_u branch remains an important computational challenge that would further validate our theoretical predictions. The methodological advances developed here, particularly the novel combination of bifurcation techniques with nonlinear boundary analysis, have broad applicability beyond ecological modeling, including in chemotaxis systems and spatial epidemiology where boundary phenomena play crucial regulatory roles. This work therefore advances both the mathematical theory of pattern formation and our fundamental understanding of how edge effects shape biological systems.

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